



Generative AI tools & Prompt Engineering

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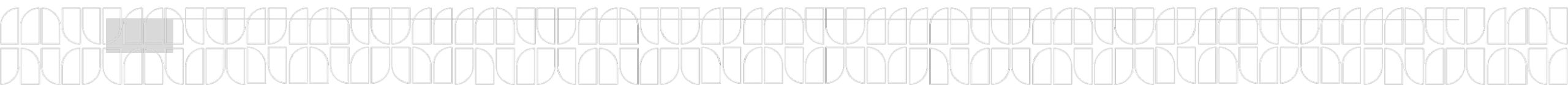
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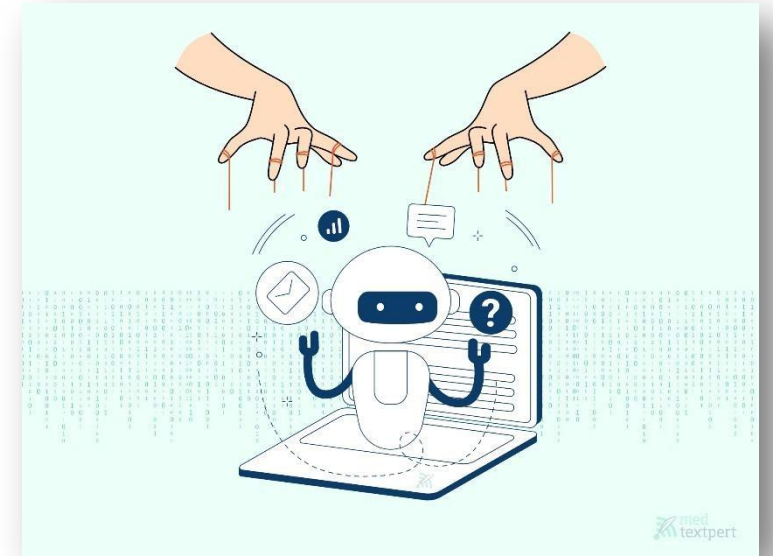
Generative AI Prompt Engineering



Generative AI Prompt Engineering



- Generative AI creates new content like text, images, and music from data.
- It learns patterns from huge datasets to produce human-like output.
- It's revolutionizing fields from education to entertainment to software development.
- Generative AI is not just smart—it's creative.
- With the right prompt, Generative AI can write, draw, or even code.



Prompt Engineering and Beyond: It's the Human Touch that Powers Generative AI Success



Generative AI

Prompt Engineering



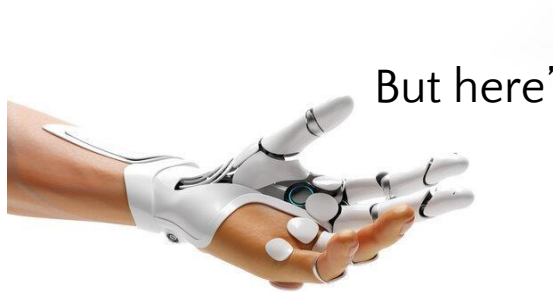
Myth	Reality
"Generative AI is always correct."	AI can confidently generate false or misleading content.
"It understands like a human."	AI predicts patterns; it doesn't truly comprehend meaning or context.
"GenAI can think or feel."	AI has no emotions, consciousness, or self-awareness.
"It creates completely original content."	It recombines existing data in new ways—not always truly creative.
"Anyone can use GenAI without skill."	Without good prompts, results can be vague or unhelpful.
"It will replace all human jobs."	GenAI supports humans—it doesn't replace creativity, empathy, or judgment.
"AI-generated content is always safe to use."	It may include bias, inappropriate, or copyrighted information.
"Only tech experts can use it."	Today's tools are user-friendly for writers, students, designers, and more.





Generative AI

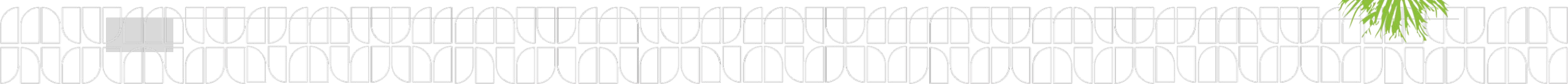
AI is evolving FAST, and many people fear that it will replace human workers.



But here's the truth they won't tell you:

AI isn't replacing **JOBS**—it's replacing **TASKS**.

And those who adapt and upskill will be the ones who THRIVE in the AI era.



About the Course

- **Course Title:** Intro to Generative AI and prompt engineering
- **Tracks:** All tracks
- **Duration:** 6 hours (Lecture + workshop)
- **Evaluation:** Attendance + presentation.
- **Prerequisites:** No prerequisites.

Resources – MaharaTech talks



1.Exploring Generative AI: Tools, Use Cases and Future Trends

<https://maharatech.gov.eg/course/view.php?id=2294>

2.Gen-AI: Tools for the Modern Developer:

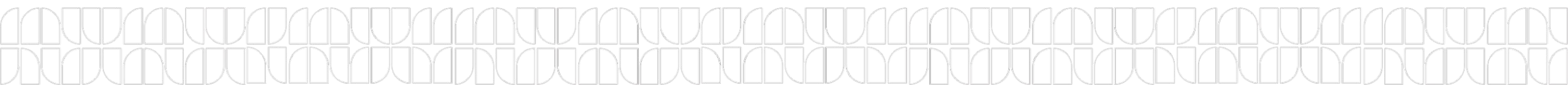
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Course Outline

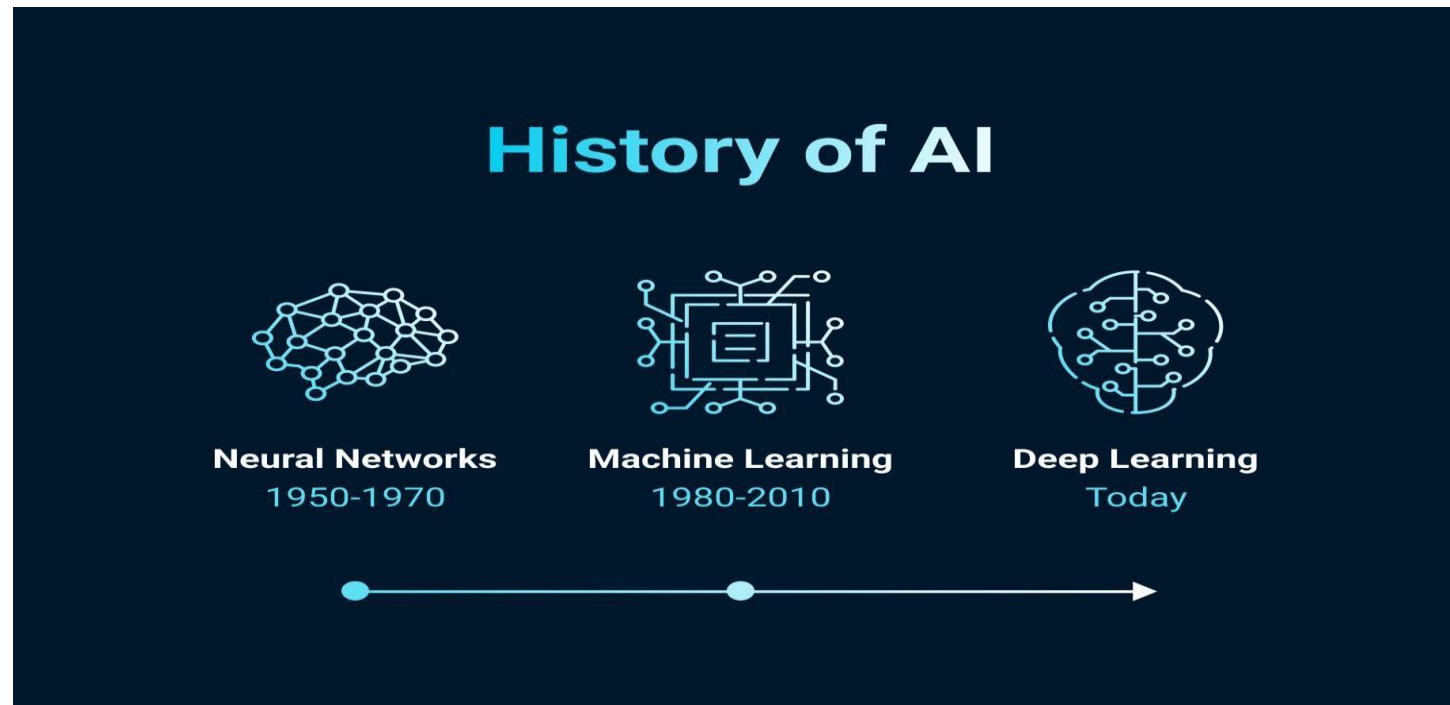


- Intro to Gen AI: evolution of AI to Generative AI.
- Gen AI benefits and practical use cases
- Key Gen AI terminologies and merging technologies: LLM, Prompts & completions, Tokens, Models, Multi-modal models, RAG, Fine-tuning, Reasoning models, AI agents.
- Gen AI tools models, tools (ChatGPT, Gemini, Claude, MS Copilot, DeepSeek, Qwen, Grok...) and use cases in different fields.
- Introducing prompt engineering
- Use cases for Gen AI applications and uses (to be tailored for each track)
- AI tools limitations, security and data privacy considerations



Why do we need to know about Gen AI?

History of AI



History of AI

The Story of AI — From Logic to Creativity

1950s – The Idea Begins

Alan Turing asked: "Can machines think?"

This question started the journey of Artificial Intelligence.

1960s–1980s – Symbolic AI

AI was built on hand-coded rules and logic.

Good at solving specific problems, but couldn't handle uncertainty or complex data.

1990s–2000s – Machine Learning

Instead of coding all rules, machines started learning from data.

This allowed better accuracy and more flexibility.

2010s – Deep Learning Revolution

Neural networks became deeper and more powerful with the rise of GPUs.

AI began understanding images, speech, and language at scale.

2020s – Generative AI (GenAI)

Now, AI doesn't just *understand* — it *creates*.



1950

Alan Turing develops the 'Turing test' which determines whether a machine can 'think' like a human.



1956

The term 'AI' is coined at a conference in Dartmouth, United States.



1968

2001, A Space Odyssey is released.



2016

The Partnership on AI is formed.



2014

Amazon's Echo is released.



2011

Apple introduces Siri as part of its iPhone 4s.



2010

The Dow Jones slumps nearly 1,000 points in the 'flash crash'.



1997

IBM's Deep Blue computer beats the reigning world chess champion, Gary Kasparov.



2016

Microsoft's Tay is released (and quickly withdrawn).



2017

It is revealed that the DeepMind NHS app test broke UK privacy law.



2018

The Cambridge Analytica scandal becomes public.



2018

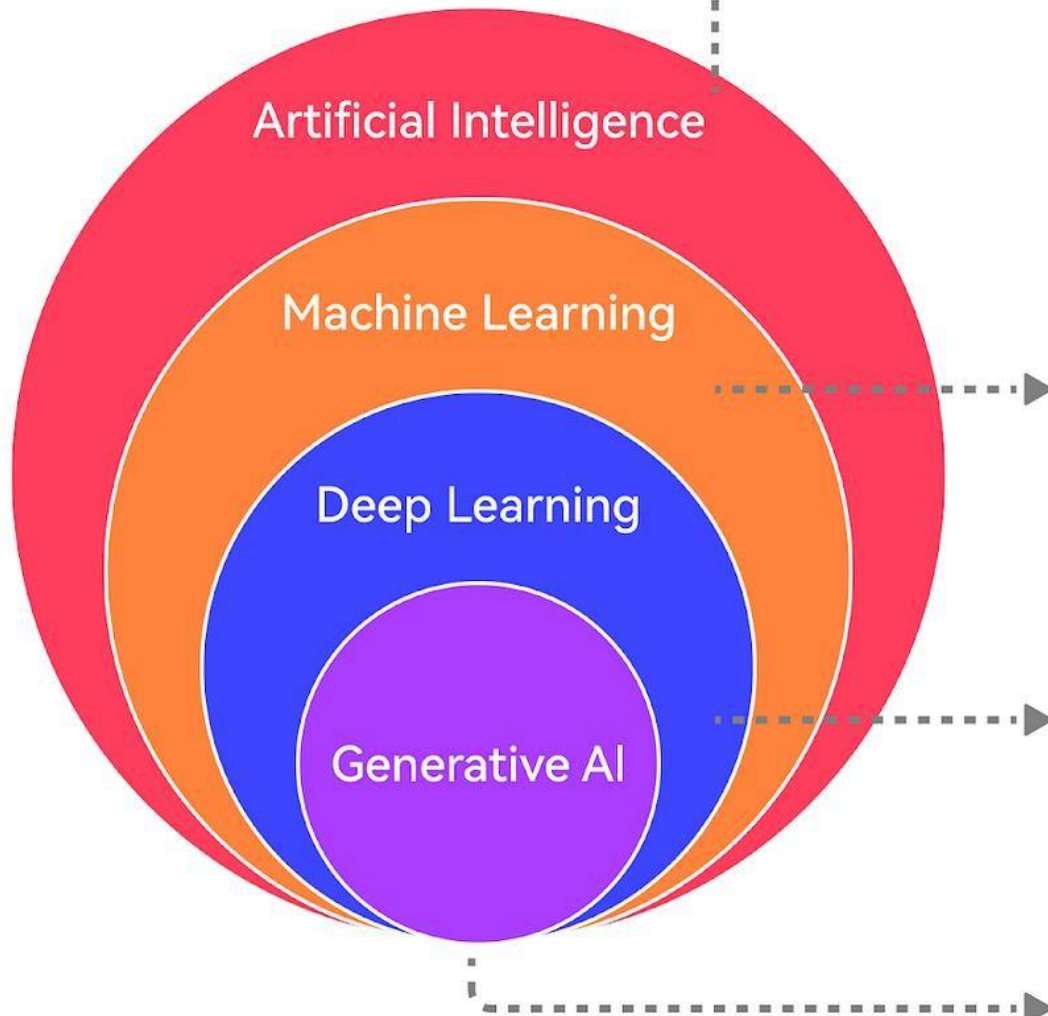
Erica, a robot, becomes a news anchor in Japan.



2019

Q, a genderless voice, is created.

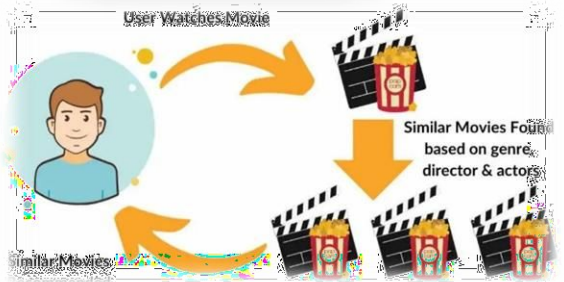




AI refers to the development of computer systems to perform tasks that need human intelligence



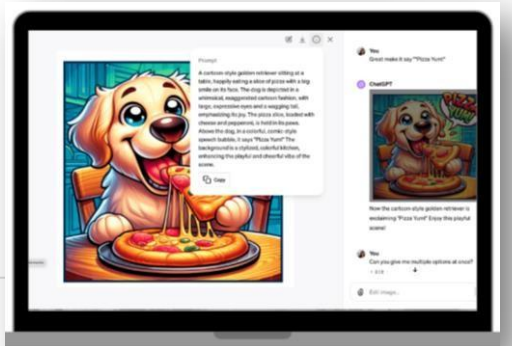
Machine learning is a subset of AI that enables computers to learn from experience



Deep learning is a subset of ML that uses neural networks for data processing



Generative AI is a subset of DL that generates content like text, images, or code based on input



What is AI, ML, Deep Learning & GenAI (with Examples)

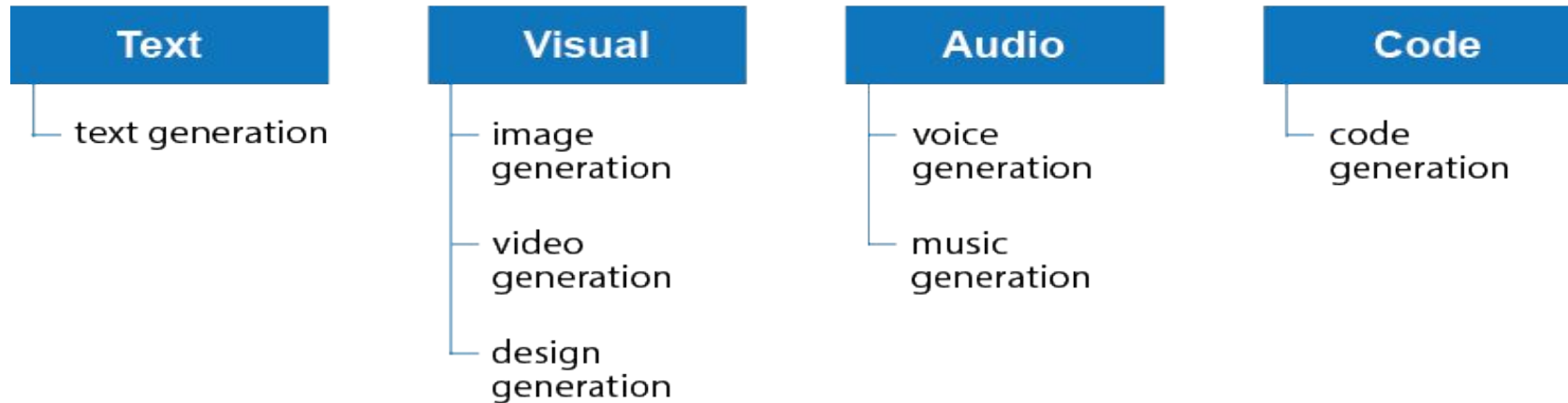
Term	Definition	Example
AI (Artificial Intelligence)	The science of making machines mimic human intelligence—like reasoning, learning, or problem-solving.	A chatbot that answers customer questions.
Machine Learning (ML)	A subset of AI where machines learn from data to improve their performance without being explicitly programmed.	A platform recommending shows based on what you watch.
Deep Learning	A subset of ML that uses neural networks with many layers to analyze complex data like images, speech, and text.	Facial recognition on your phone.
Generative AI (GenAI)	A type of AI that can create new content—like text, images, or code—based on patterns it learned from training data.	ChatGPT writing an email or DALL·E generating art from a prompt.

What's Generative AI?

Generative AI: is a type / subset of artificial intelligence system that can produce various types of content, including text, imagery, audio and synthetic data.

Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics.

Generative AI Tools



Text Generation

- ChatGPT writing essays, summaries, or poetry
- Jasper creating marketing copy

Image Generation

- DALL-E generating artwork from text prompts
- Midjourney creating stylized illustrations or concept art

Music Creation

- Aiva composing classical-style music
- Suno.ai generating songs from a theme or lyrics

Video & Animation

- Runway ML turning text into short video clips
- Pika Labs generating animated scenes from prompts

Code Generation

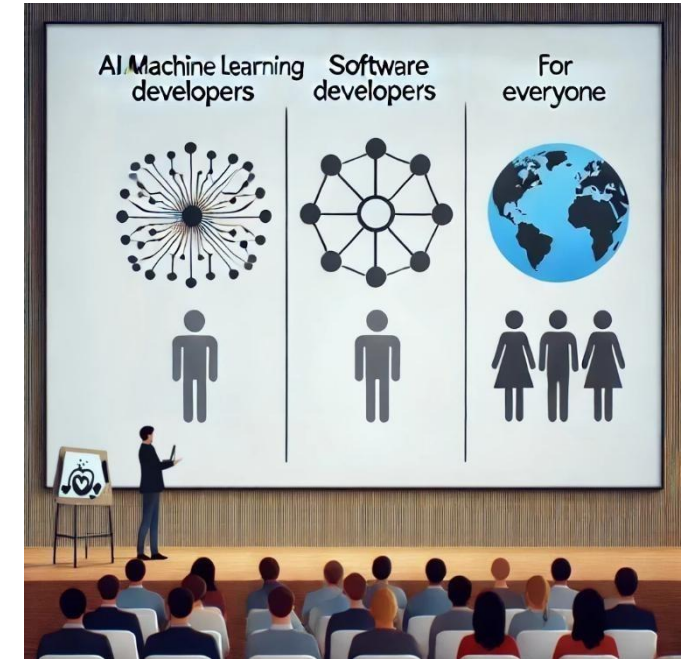
- GitHub Copilot suggesting code snippets inside editors
- Replit Ghostwriter building small applications or debugging

Speech & Voice

- ElevenLabs cloning voices or generating speech from text
- Descript Overdub editing audio with AI-generated voice

Who should learn about Gen AI?

- **AI / Machine Learning Developers:**
 - Design and refine Generative AI models to create more intelligent and efficient AI-driven solutions.
- **Software Developers:**
 - Utilize AI tools to enhance productivity in development tasks.
 - Leverage Generative AI APIs to build AI-powered applications
- **Everyone:**
 - In today's AI-centric world, it's crucial for individuals across all professions to grasp the basics of Generative AI and incorporate AI tools into their workflows to boost productivity.



Will AI Replace Jobs / employees?

“
Q: Will AI take away your job?

A: AI won't take your job, but someone using AI will.



SHASHANK SINGH

CEO, Bakstage

<https://bakstage.ai>

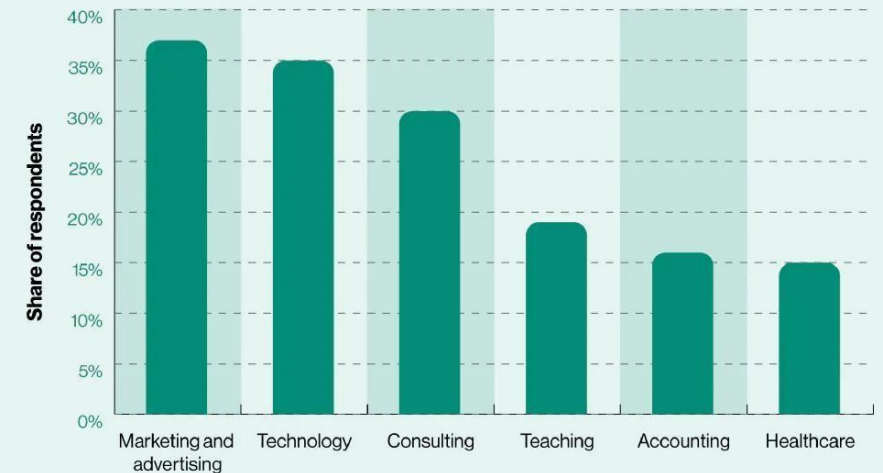


NEWS

As many as 41% of employers plan to use AI to replace roles—but it's not a 'jobs apocalypse,' experts say

By Ryan Johnston, CNBC • Published February 26, 2025 •

Rate of generative AI adoption in the workplace in the United States 2023, by industry



Source: Statista

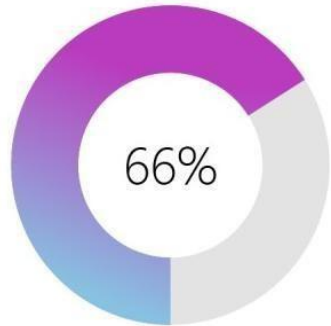
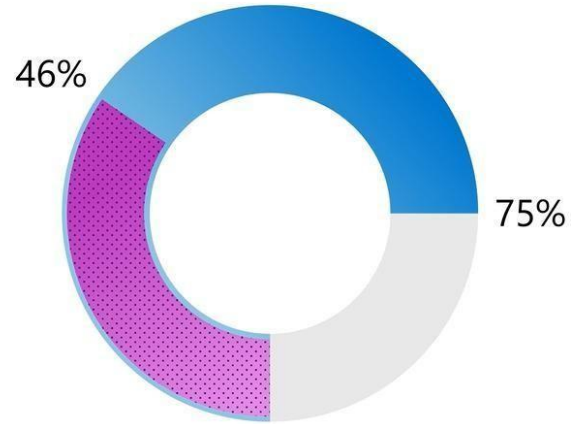
Gen-AI Tools in Work

Three Out of Four People Use AI at Work

Usage nearly doubled in the last six months.

75% of people are
already using AI at work

46% of them started using
it less than 6 months ago



Leaders would
not hire someone
without AI skills

AI-Human productivity gains in selected professions

Time to complete coding tasks in
minutes 2022

Time and cost to complete design
tasks



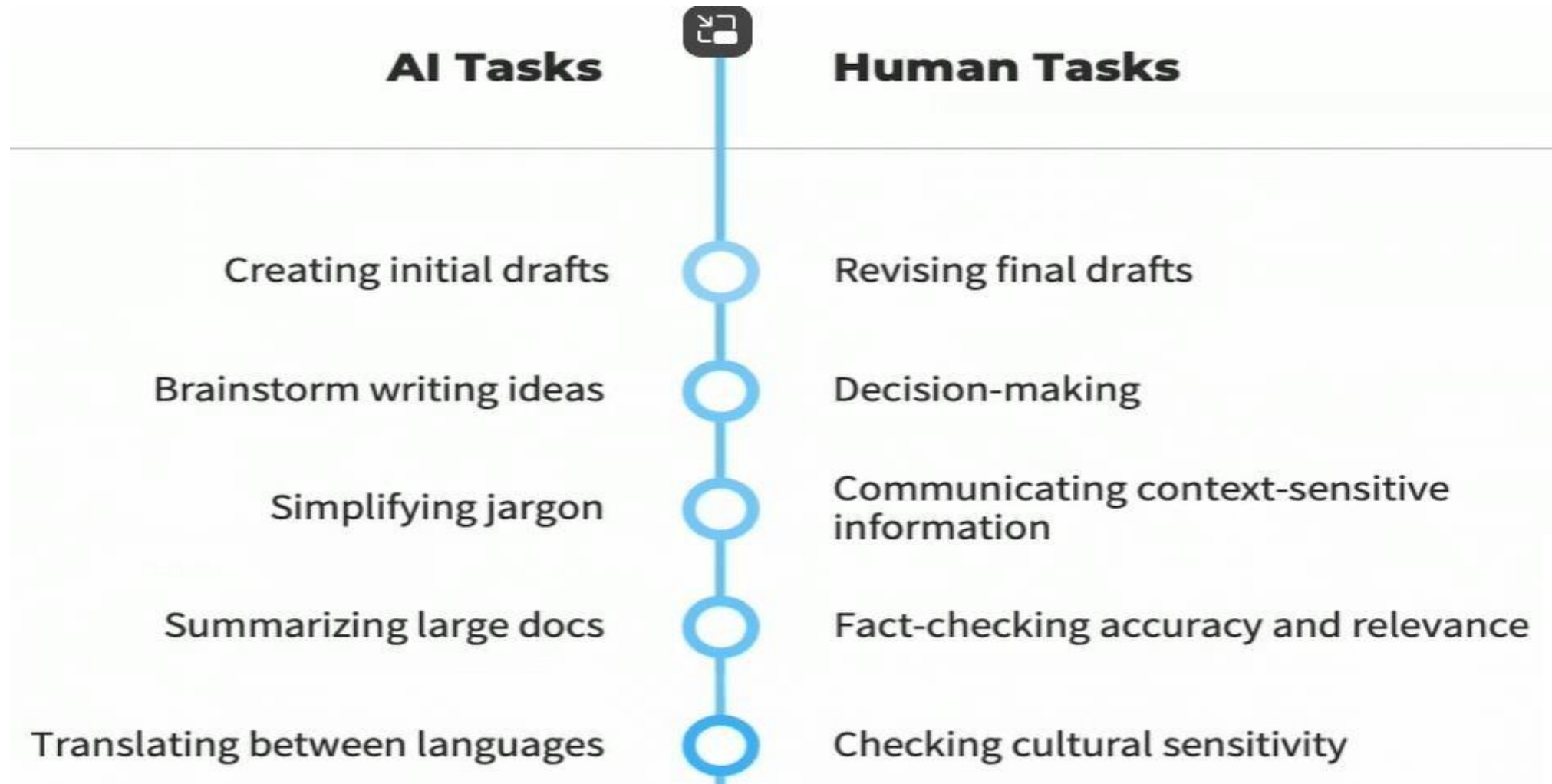
Human
Cost: \$150
Time: 5 hours



**Generative
AI**
Cost: \$0.08
Time: <1
minute

2024 Work Trend Index Annual Report from Microsoft and LinkedIn:
AI In Work

Tasks



Data Center GPUs

- High-performance graphics processors (e.g., NVIDIA A100, H100)
- Enable massive parallel processing for training and running GenAI models
- Hosted in cloud data centers (like AWS, Google Cloud, Azure)

AI Models

- Pre-trained neural networks that can understand and generate content
- Examples: GPT (OpenAI), BERT (Google), Stable Diffusion (Stability AI)



Model Platforms

- Tools to train, fine-tune, and deploy models
- Examples: Hugging Face, TensorFlow, PyTorch, NVIDIA NeMo

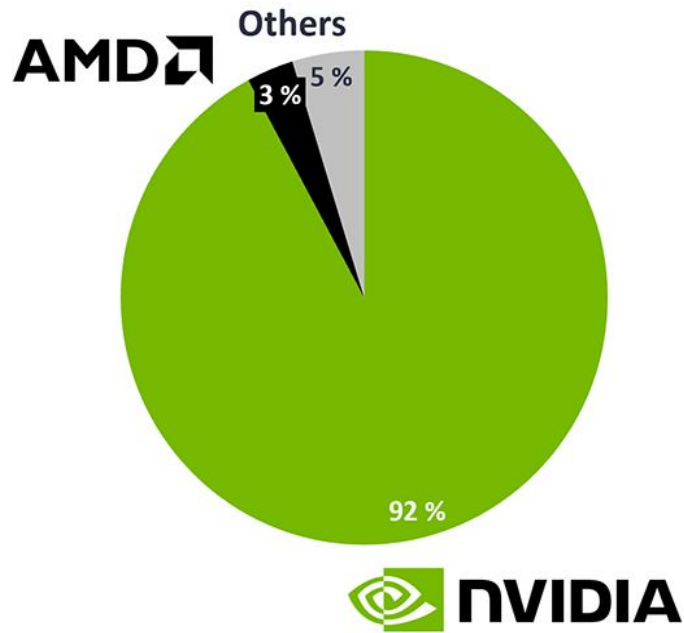
AI Services

- Ready-to-use APIs or tools for users and developers
- Examples: OpenAI API (ChatGPT), Google Vertex AI, Amazon Bedrock, Microsoft Azure OpenAI

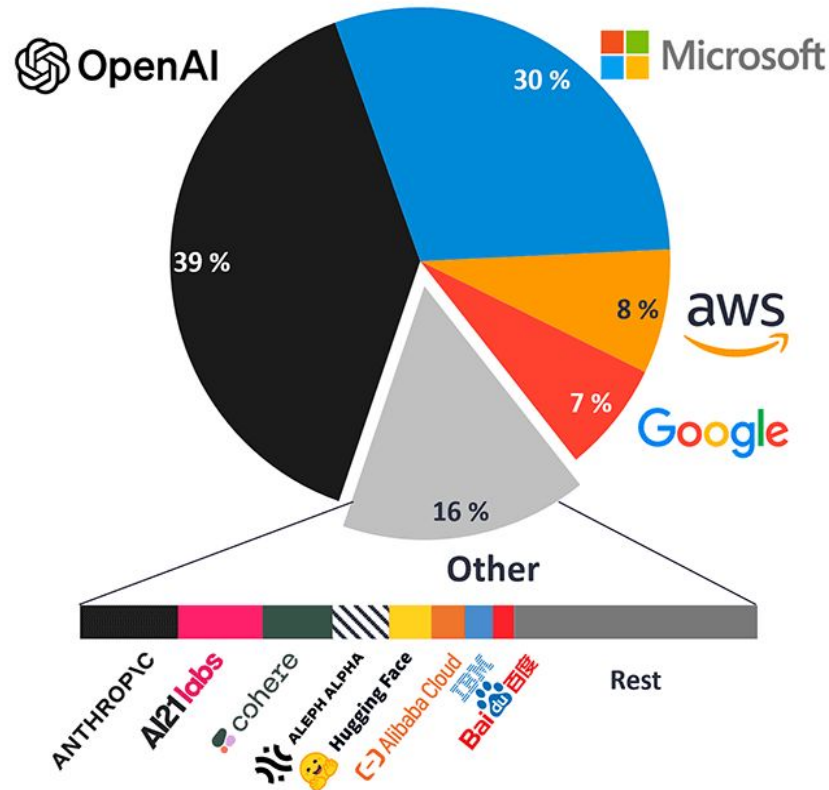


Generative AI: Market share of leading vendors 2023

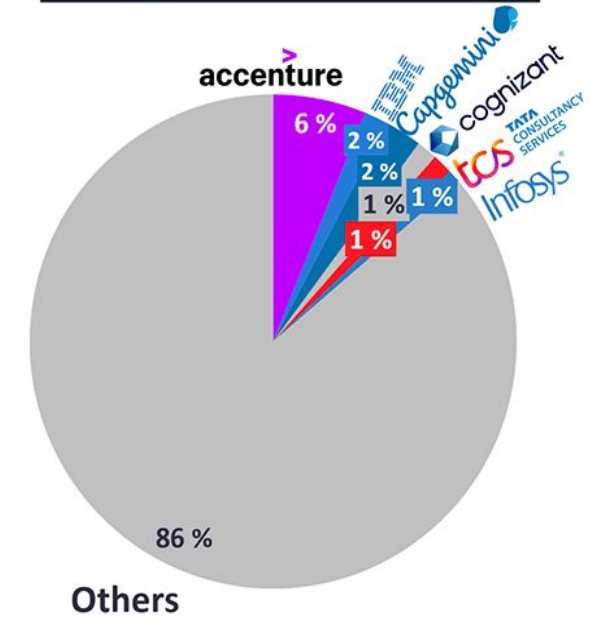
1 Data center GPUs



2 Models & platforms



3 Services



Note: Numbers are rounded and might not add up to 100%; Market share is based on 2023 market sizes (based on revenue).

Source: IoT Analytics Research 2023-Generative AI Market Report 2023-2030. We welcome republishing of images but ask for source citation with a link to the original post and company website.

Sample GenAI Freelancing jobs



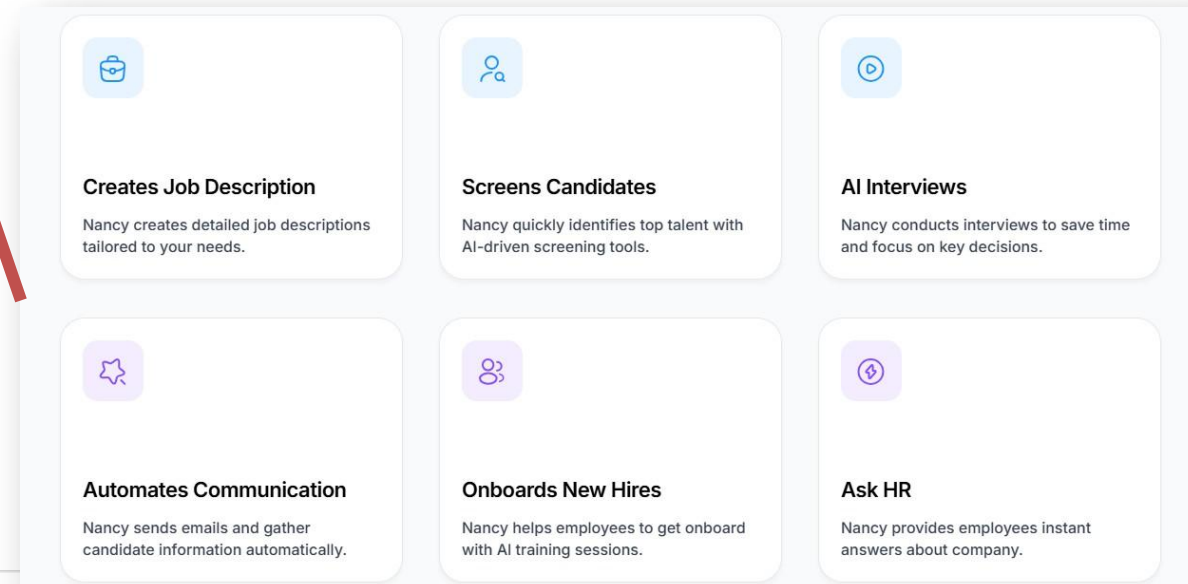
- **Upwork**
 - [AI Image Generation and Deepfake Model Creation](#)
 - [AI developer \(rag + llm\) + fullstack](#)
 - [Next.js Developer Needed for SaaS Application Integration with AI Features](#)
 - [Developers for 5 AI agents](#)
 - [AI Engineer for Medical RAG-based Chatbot Development](#)
 - [GenAI engineer to build GenAI-powered modules of an astrology app](#)
 - [CRate AI Agent using Open AI or Google AI API or CrewAI](#)
 - [Full-Stack Developer with AI Experience \(Replit Preferred\)](#)
 - [Developer Needed to Build AI Agents for Data Analysis and Digital Advertising](#)
 - [Web App Development with AI for Meta Ads Management](#)
 - [... and more](#)
- **Khamsat and Mostaqi**
 - [Khamsat Gen AI Services](#)
 - [Mostaqi Gen AI Projects](#)



Gen-AI based startups and real products

- Use cases for real products using Gen AI:

- Loop-x (AI Agents for different roles): <https://loop-x.co>
- Gemelo ai (Twin AI creator): <https://gemelo.ai>
- Apriora AI (Recruitment and Interviews): <https://www.apriora.ai>
- Velents AI (Recruitment and Interviews): : <https://www.velents.com>
- Nancy AI (Recruitment and Interviews): : <https://nancy-ai.com>
- Mariana AI (Medical): <https://marianaai.com>



<https://www.facebook.com/reel/605938575420991>

How to use the AI tools responsibly?

- **As a Student/Learner:**

-  **Use AI for:** Asking for explanations, self-testing, researching, and similar activities.
-  **Don't use AI for:** Assignments, projects, tasks, or topics you're still learning.
- **Build your skills** to work effectively both with and without AI.



- **In a Job or Freelancing:**

- **Check first** if AI tool usage is permitted and which tools are allowed.
- **Use AI to boost productivity** and improve your work.
- **Always review** AI-generated content carefully.
- **Don't rely on AI** for tasks you have no prior knowledge of.

Introducing Generative AI

<https://www.apriora.ai/>

<https://www.youtube.com/watch?v=Q7dqyoHUI5I>

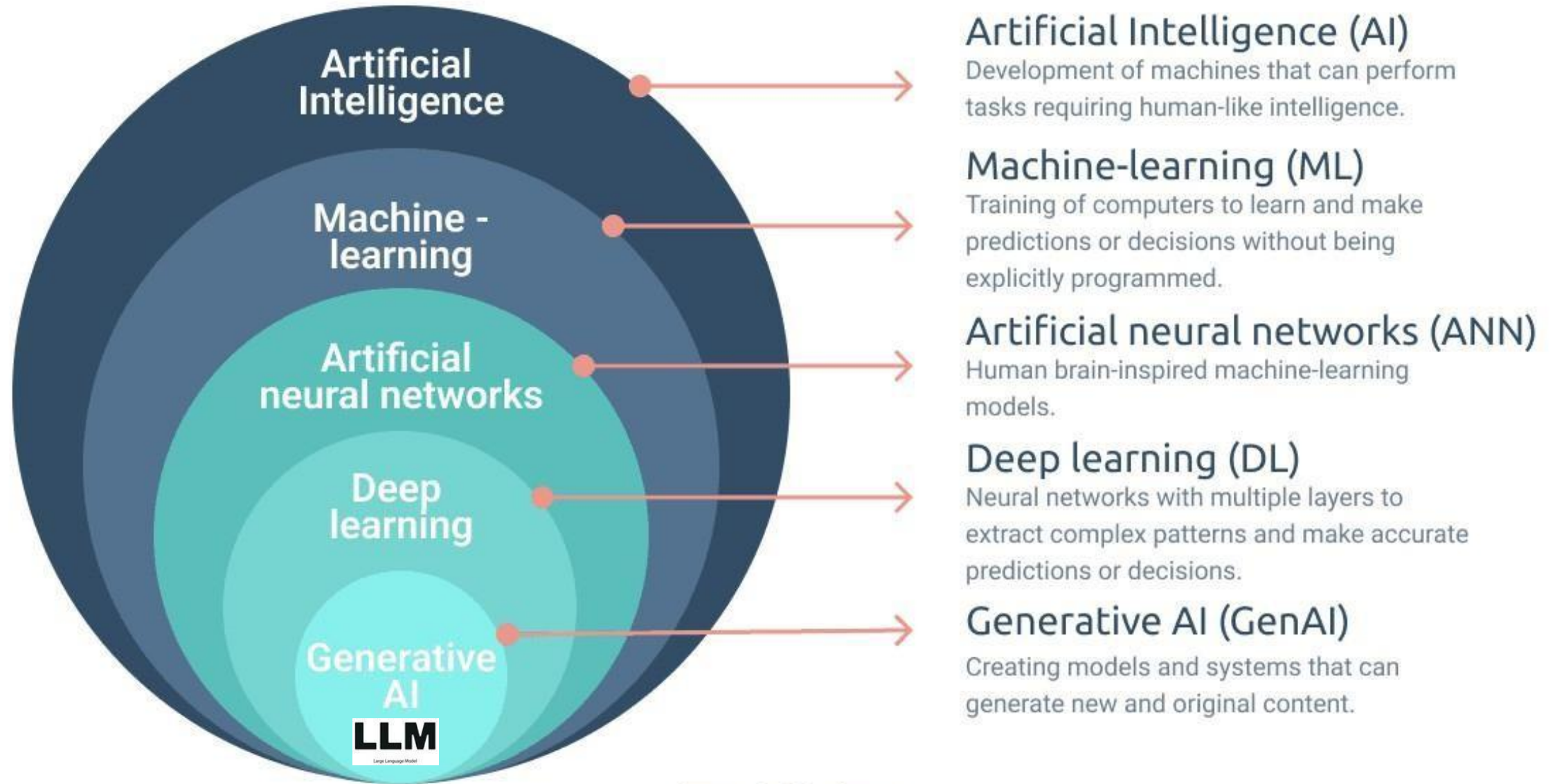


Image 2. AI subsets

AI terms

- **A Large Language Model (LLM):**

- Is an advanced AI system trained to understand and generate human-like text.
- It learns from massive amounts of text data and can answer questions, write content, and assist in conversations.

Large Language Models (LLMs)

- **Definition:** A subset of Generative AI trained on massive text data to understand and generate human-like language.
- **Scope:** Focused on natural language tasks (chatting, summarizing, translating, etc.)
- **Purpose:** To process and generate text using deep learning (usually transformers).

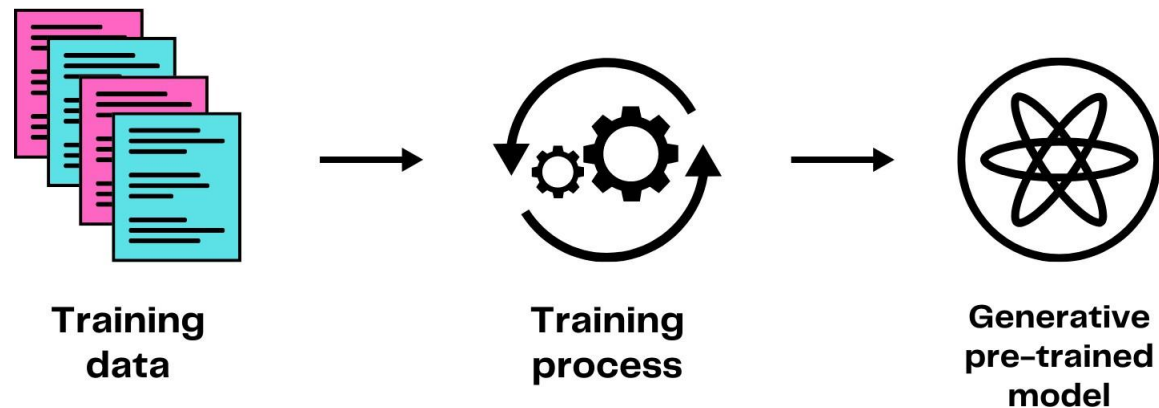
Examples:

- GPT-4 (OpenAI)
- LLaMA (Meta)
- Claude (Anthropic)



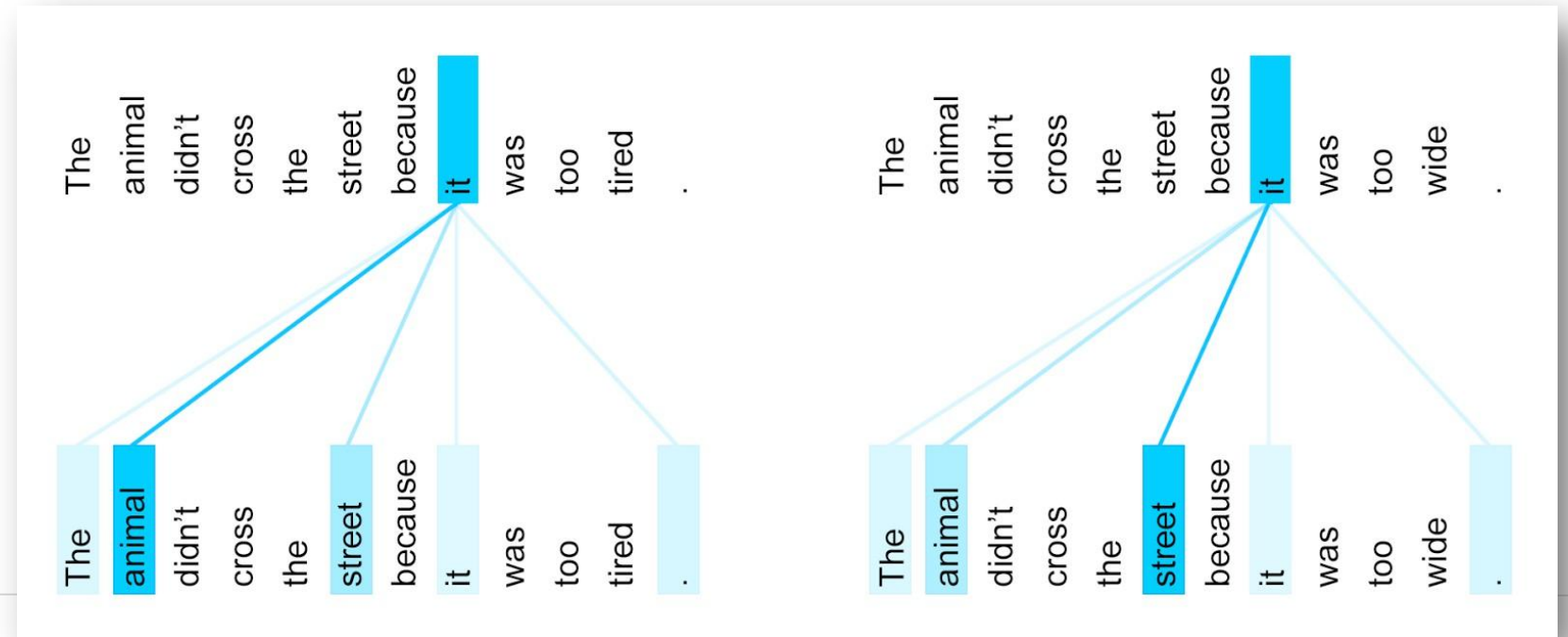
AI terms

- **AI model:** is a system trained to recognize patterns, make predictions, and automate tasks based on data. It learns from examples and improves its performance over time..
- **Multimodal models:** can process a wide variety of inputs, including text, images, and audio, as prompts and convert those prompts into various outputs, not just the source type.
- **Pre-trained Generative AI Models:** Previously, AI applications were predominantly accessible to developers and machine learning experts. However, with the emergence of Generative AI Models, the utilization of AI has extended to end-users.
- Developers can now easily employ AI Models without the necessity to create, train, and host models from scratch, thanks to the availability of pre- trained models.



AI terms

- **Reasoning Model** is a type of AI system designed to simulate human logical thinking. It processes information through a set of rules or logic to draw conclusions, solve problems, or make decisions.
- **Deep Thinking** is the ability to analyze complex problems, explore multiple perspectives, and make well-reasoned decisions. It involves **critical analysis, logical reasoning, and reflection** rather than relying on surface-level understanding.
- **Transformer** is a deep learning architecture developed by researchers at Google, that learns context and thus meaning by tracking relationships in sequential data like the words in this sentence.



AI terms (Cont.)

- **Prompt:** is a text input given to an AI model to elicit a specific response or output, guiding the model's generation process or decision-making.
- **Completion:** is the output or response generated by an AI model based on a given prompt, representing the model's attempt to complete, answer, or extend the input it received.
- **Tokens:** For text tokens Azure OpenAI processes text by breaking it down into tokens. For example: 1,000 tokens is about 750 words.
- **Prompt engineering** is the process of structuring an instruction that can be interpreted and understood by a generative AI model. A prompt is natural language text describing the task that an AI should perform.

AI terms (Cont.)

💬 Prompt

- The input or instruction given to a generative AI model.
- Example: "Write a bedtime story about a robot and a cat."

🔥 Completion

- The AI-generated response to a prompt.
- Example: "Once upon a time, a robot named Zeno found a lost cat in a scrapyard..."

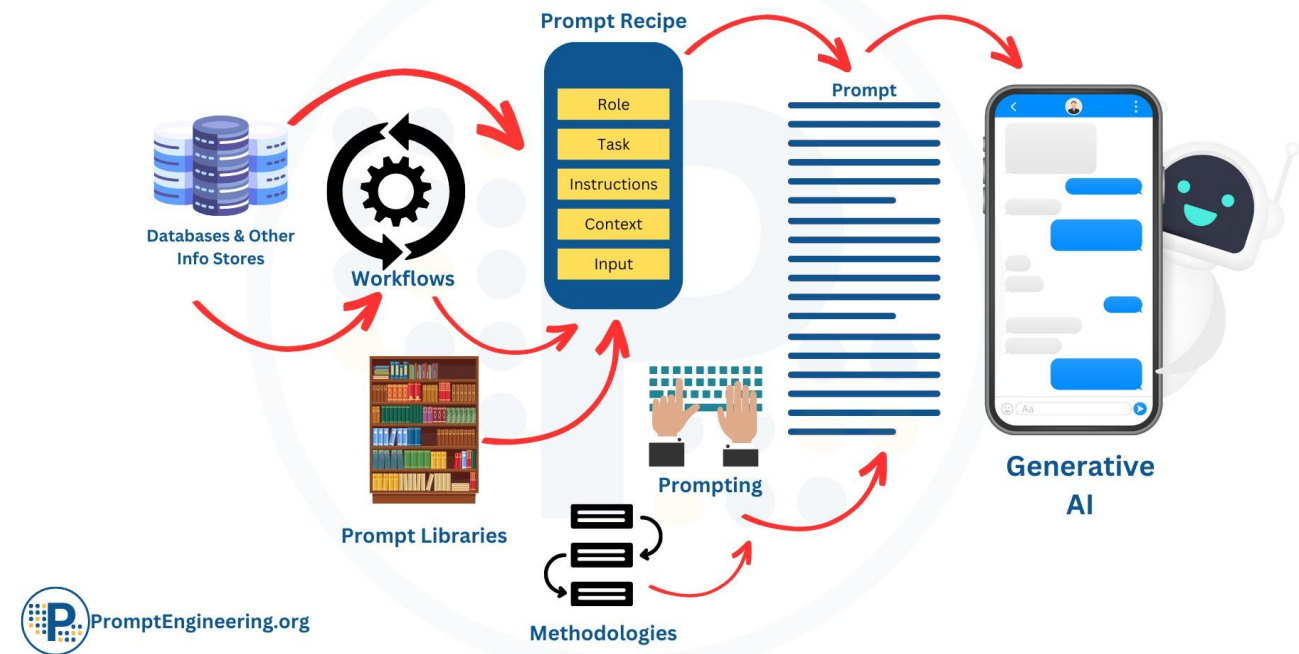
📄 Tokens

- Units of text (words or parts of words) used by the model to process input/output.
- Example:
 - "ChatGPT is smart" = 4 tokens
 - Most English words ≈ 1-2 tokens

🛠️ Prompt Engineering

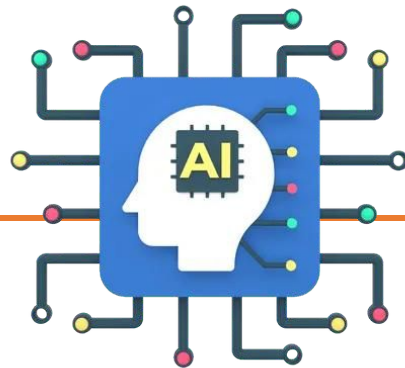
- The skill of crafting clear, specific, and effective prompts to get the best results from AI.
- Example:
 - ❌ Bad: "Tell me something."
 - ✅ Good: "Give me 3 pros and cons of online learning for university students."

What is Prompt Engineering? Everything that goes before the prompt



It's the Human Touch that Powers Generative AI Success

Models and Tools





is an AI research and deployment company focused on ensuring that artificial general intelligence (AGI) benefits all of humanity.

OpenAI Models Overview

Model	What It Does	Example Use
GPT	Generates text, answers questions, writes code	Chatbots, content creation
DALL-E	Creates images from text prompts	AI art, concept visuals
Whisper	Converts speech to text	Transcriptions, subtitles
Codex	Turns text into code	GitHub Copilot, coding help
Sora	(In development) Generates videos from text	Creative video concepts

GPT stands for **Generative Pre-trained Transformer**.

- **Generative:** It creates (generates) text.
- **Pre-trained:** It's trained on a large dataset before fine-tuning.
- **Transformer:** The neural network architecture it's based on.

OpenAI Models

[Models - OpenAI API](#)

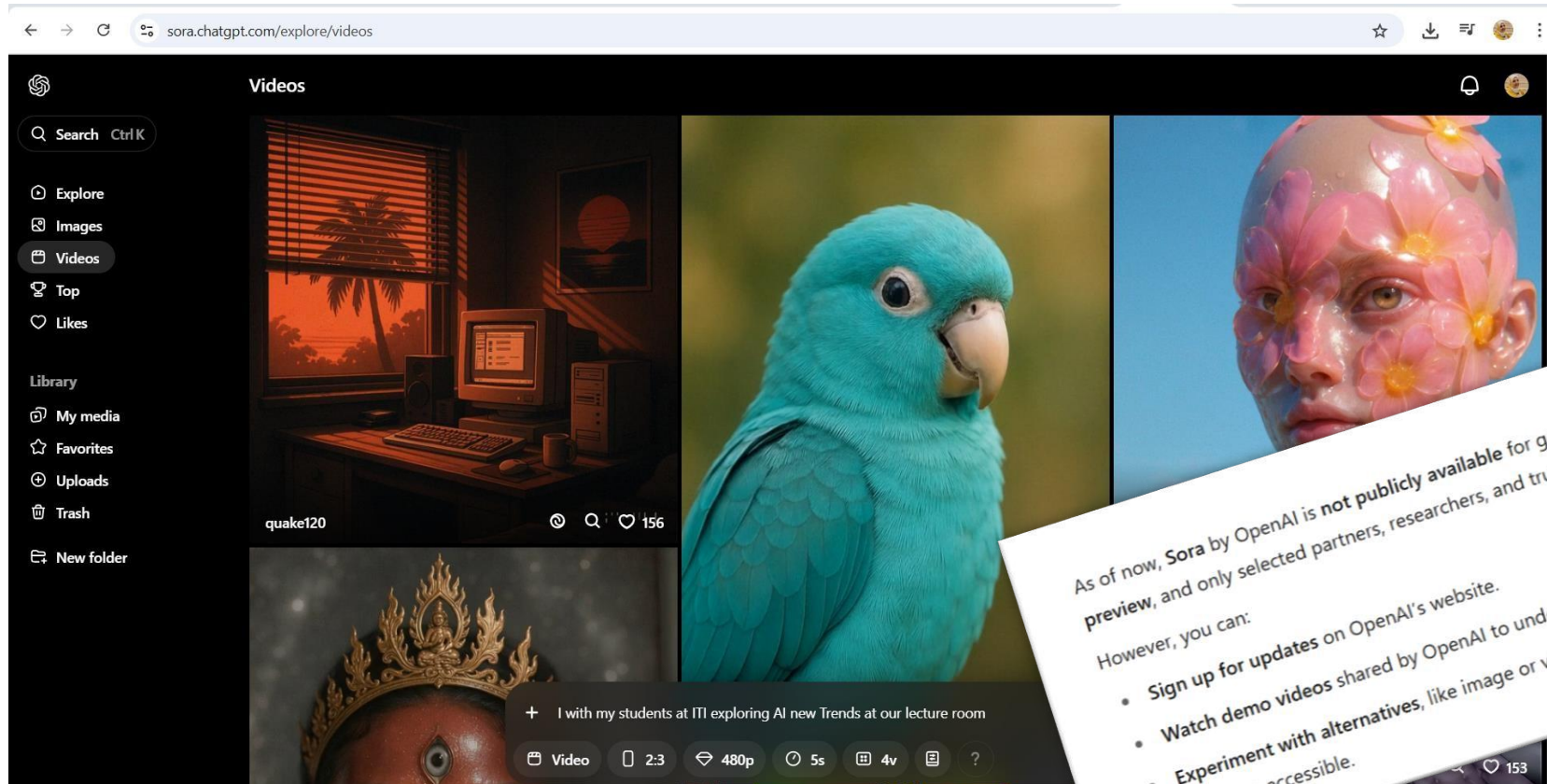
Models overview

The OpenAI API is powered by a diverse set of models with different capabilities and price points. You can also make customizations to our models for your specific use case with [fine-tuning](#).

MODEL	DESCRIPTION
GPT-4o	Our high-intelligence flagship model for complex, multi-step tasks
GPT-4o mini	Our affordable and intelligent small model for fast, lightweight tasks
o1-preview and o1-mini	Language models trained with reinforcement learning to perform complex reasoning.
GPT-4 Turbo and GPT-4	The previous set of high-intelligence models
GPT-3.5 Turbo	A fast, inexpensive model for simple tasks
DALL·E	A model that can generate and edit images given a natural language prompt
TTS	A set of models that can convert text into natural sounding spoken audio
Whisper	A model that can convert audio into text
Embeddings	A set of models that can convert text into a numerical form
Moderation	A fine-tuned model that can detect whether text may be sensitive or unsafe

Sora: Text-to-video model

- **Sora:** OpenAI's text-to-video model, it is an AI model that can create realistic and imaginative scenes from text instructions. <https://openai.com/sora/>



As of now, Sora by OpenAI is **not publicly available** for general use or free access. It is still in **research preview**, and only selected partners, researchers, and trusted testers may get early access.

However, you can:

- Sign up for updates on OpenAI's website.
- Watch demo videos shared by OpenAI to understand its capabilities.
- Experiment with alternatives, like image or video generation tools (e.g., DALL-E, Runway, Pika Labs) that are more accessible.

GPT-o1: OpenAI's New Reasoning Model

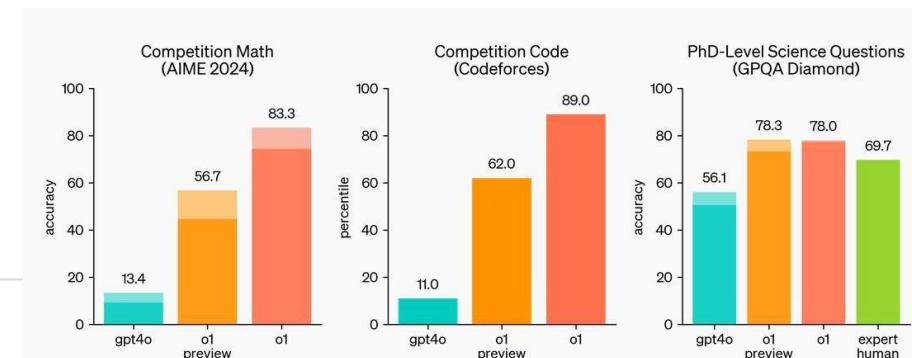
What is [GPT-o1](#)?

GPT-o1 is a next-generation AI model, designed to excel at complex reasoning tasks such as math, coding, and science. Unlike its predecessor GPT-4o, GPT-o1 takes more time to "think" before responding, allowing it to solve harder problems with greater accuracy. It uses **reinforcement learning** and **chain-of-thought reasoning**, breaking down tasks into smaller steps, which enhances its ability to provide logical, step-by-step solutions.

OpenAI o1 ranks in the **89th percentile** on competitive programming questions (Codeforces), places among the top 500 students in the US in a qualifier for the USA Math Olympiad (AIME) and **exceeds human PhD-level** accuracy on a benchmark of physics, biology, and chemistry problems (GPQA).

Chain of Thought

Similar to how a human may think for a long time before responding to a difficult question, o1 uses a chain of thought when attempting to solve a problem. Through reinforcement learning, o1 learns to hone its chain of thought and refine the strategies it uses. It learns to recognize and correct its mistakes. It learns to break down tricky steps into simpler ones. It learns to try a different approach when the current one isn't working. This process dramatically improves the model's ability to reason.



Top AI models and Chat-based AI tools



AI tools Capabilities

AI tools capabilities and use cases:

- **Text generation:** can generate text based on a given prompt, including full sentences, paragraphs, and even articles.
- **Conversational AI:** can be used to build chatbots and conversational interfaces, allowing it to handle various types of user inputs, such as natural language questions and commands.
- **Sentiment Analysis:** can determine the sentiment of a given text, whether it is positive, negative, or neutral.
- **Translation:** can translate text from one language to another, supporting multiple languages.
- **Question Answering:** can provide answers to questions asked in natural language.
- **Content creation:** can be used to generate content, such as articles, stories, and poems.
- **Text Classification:** can classify text into different categories, such as topics, genres, and sentiments.
- **Code Generation:** can generate code snippets based on given natural language descriptions.
- **And more...**

ChatGPT features and use cases

Exploring
ChatGPT –
as one of
the most
known
tools –
features:

Image generation and Multimodal capabilities.

Data analytics

ChatGPT Memory

Custom Instructions

Custom GPTs

Code interpreter and generating files

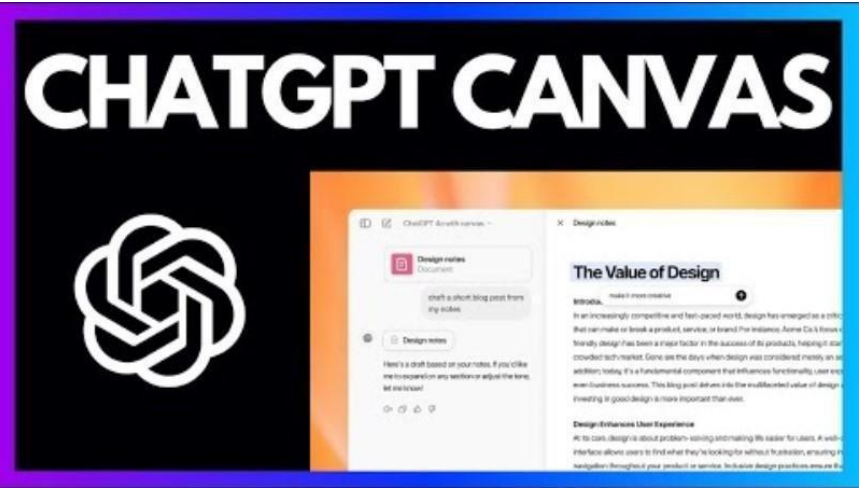
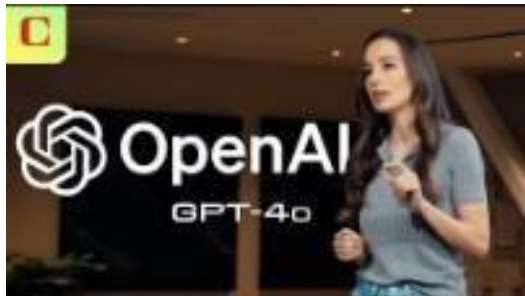
Web browsing

Advanced voice mode

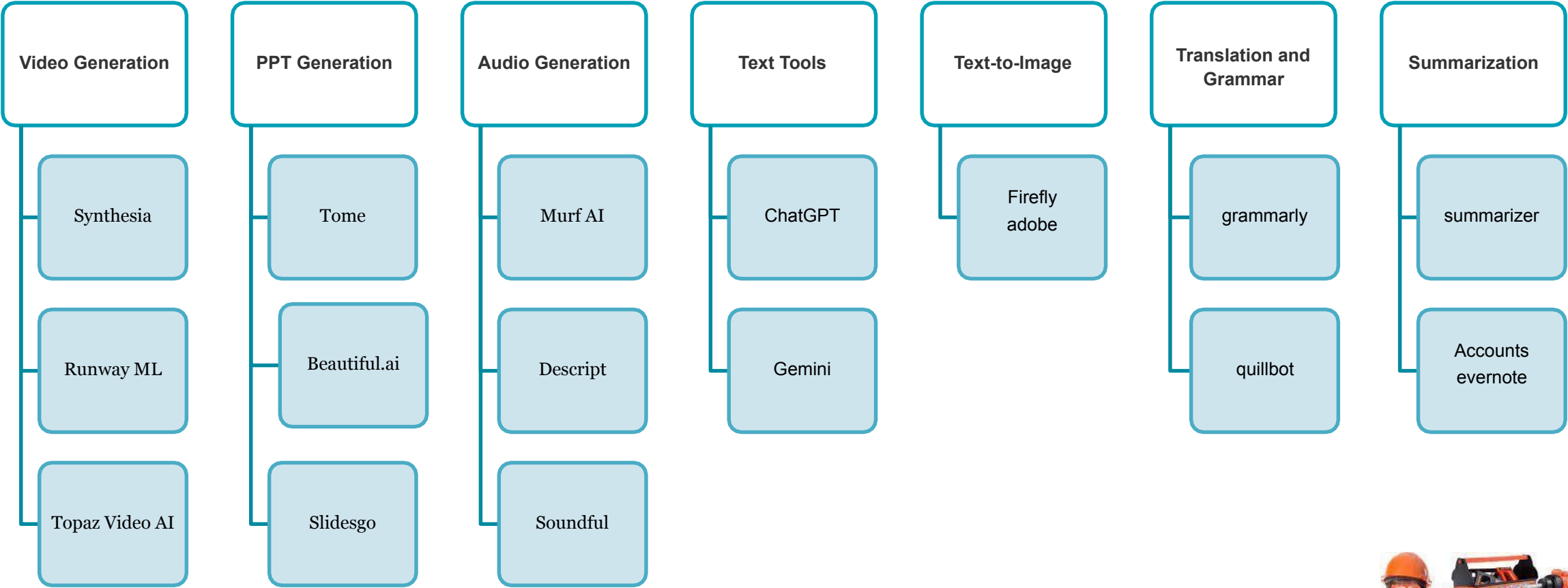
GPT o1 Models

GPT-4o with canvas

Exploring Gen-AI tools new features



Exploring Gen-AI tools new features



Prompt Engineering



Introducing prompt engineering

Prompt engineering:

- Prompt Engineering is the art and science of crafting inputs (**prompts**) that guide an AI, especially a language model, to produce desired outputs or responses.
- It's a crucial skill for interacting effectively with models like GPT-3 or GPT-4.

Why It Matters?

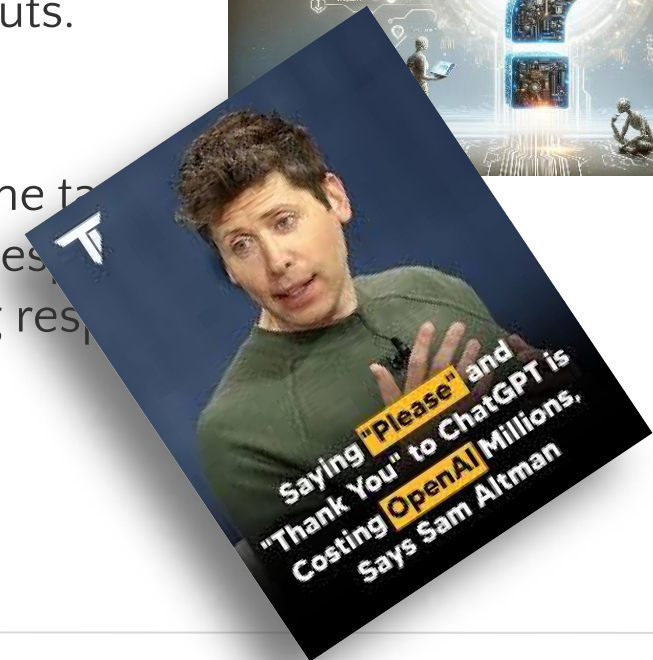
- The quality of the prompt directly influences the AI's response.
- Good prompts lead to more accurate, relevant, and useful outputs.

Key Elements:

- **Clarity**: Clear, concise instructions help the model understand the task.
- **Context**: Providing relevant background information enhances responses.
- **Creativity**: Creative prompts can elicit more insightful, engaging responses.

Challenges:

- Balancing specificity and openness in prompts.
- Avoiding biases and ensuring ethical AI interactions.



Elements of a Prompt

A prompt contains any of the following elements:

- **Instruction** – a specific task or instruction you want the model to perform
- **Context** – external information or additional context that can steer the model to better responses
- **Input Data** – the input or question that we are interested to find a response for
- **Output Indicator** – the type or format of the output.

Sample Prompt:

- **Instruction:** Summarize the article.
- **Context:** The article is from a scientific journal, and it discusses the latest advancements in renewable energy technology.
- **Input Data:** [Insert link to or text of the article here]
- **Output Indicator:** Provide a concise summary in bullet points, focusing on key findings and technological innovations.



Prompt Engineering

Design & Structure

Prompt Engineering's Design & Structure

- 1- Subject Focused Prompt
- 2- Problem Focused Prompt
- 3- Learner Focused Prompt

Prompt Engineering's Design & Structure

1- Subject Focused Prompt

Problem is Focused on specific topic or domain of knowledge.

Use Case :

When user wants focused, detailed information or summary about specific topic.

Example:

Explain the concept of inheritance in object – oriented programming using javascript.

Prompt Engineering's Design & Structure

1- Subject Focused Prompt

Problem is Focused on specific topic or domain of knowledge.

Usually Starts with:

- “Explain ...”
- “Describe ...”
- “Discuss ...”
- “What is ...”
- “How does ... work?”

Prompt Engineering's Design & Structure

Problem Focused Prompt

Examples:

How can I fix a coding Problem ?

How can I fix a Null Pointer Exception Error in C# ?



TRUE



FALSE

Prompt Engineering's Design & Structure

2– Problem Focused Prompt

The prompt is framed around a problem that needs solving. It presents a challenge or question where the AI helps by providing solutions, strategies, or explanations.

Use Case :

Useful for coding issues, debugging, troubleshooting, or decision – making tasks.

Prompt Engineering's Design & Structure

2- Problem Focused Prompt

Common question:

- “How can we solve...?”
- “What can be done about...?”
- “What are some ways to improve...?”
- “Identify causes and propose solutions for...”
- “Suggest a plan to address...”

Example:

- How can I fix a Null Pointer Exception Error in C# ?
- I feel tired even after getting 8 hours of sleep. What factors should I consider before seeing a doctor?
- I want to switch to a healthier diet without spending too much money. Can you help me plan affordable meal options?

Prompt Engineering's Design & Structure

3- Learner Focused Prompt

The Prompt is tailored to the learner's level, background, and needs. It adapts content delivery to match the learning style or pace of the person using the AI.

Goal : Support Personalized learning, scaffolding, and interactive teaching.

Use Case :

Ideal for education, tutoring, onboarding beginners, or creating step-by-step guides.

Prompt Engineering's Design & Structure

3- Learner Focused Prompt

Often Starts With:

- “Explain in your own words...”
- “Reflect on...”
- “How would you apply...”
- “What did you find most challenging about...”
- “Describe a time when...”

Example:

- I'm new to java programming. Can you explain what variable is with simple examples ?
- If $2x + 3 = 7$, what do you think the first step might be to get x by itself?

Prompt Engineering

Techniques & Approaches

Prompt Engineering's Techniques & Approaches

- 1- Zero-shot prompting
- 2- One-shot prompting
- 3- Few-shot prompting
- 4- Chain of Thought prompting
- 5- RTF prompting
- 6- RISEN prompting

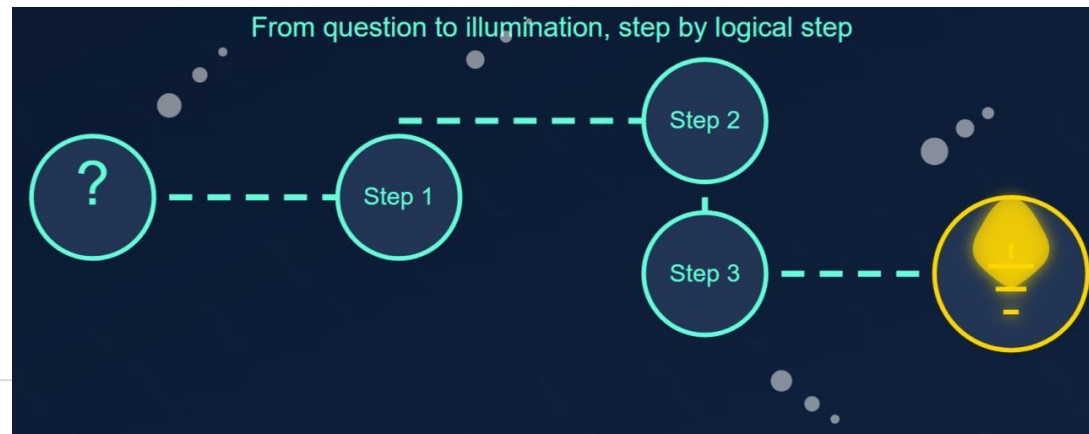
Prompt Engineering techniques and approaches [CoT]

CoT (Chain of Thought):

- It involves structuring prompts in a way that leads the AI through a step-by-step reasoning process.
- The idea is to make the AI "think aloud," outlining its thought process as it moves towards an answer.
- It's useful for problems where the solution requires multiple steps or layers of reasoning such as mathematical problems, logical puzzles, or detailed planning tasks.

Prompt template:

"Explain how you would solve [describe the problem or question]. Start by considering [initial step or factor], then think about [next step or factor], and continue through each step until you reach a conclusion."



Prompt Engineering techniques and approaches [CoT]

CoT (Chain of Thought):

Examples:

"How to increase our sales of our new smart phones. "



TRUE



FALSE

"Outline the steps you would take to increase sales of a new smartphone. Begin by identifying the target market for the smartphone. Then, think about the key features that would appeal to this market. Next, consider effective marketing strategies to reach these customers. Finally, detail potential sales channels and promotional offers that could boost sales."

Prompt Engineering techniques and approaches [CoT]

CoT (Chain of Thought):

Examples:

"Outline the steps you would take to increase sales of a new smartphone. Begin by identifying the target market for the smartphone. Then, think about the key features that would appeal to this market. Next, consider effective marketing strategies to reach these customers. Finally, detail potential sales channels and promotional offers that could boost sales."



Prompt Engineering techniques and approaches [Zero-shot prompting]



Zero-shot prompting:

- It involves presenting a task to the AI model without any prior specific examples or training for that task.
- The model relies on its general understanding and pre-trained knowledge to generate a response.

✓ When to Use Zero-shot Prompting

1. **Simple Tasks**
→ Like summarizing, translating, or generating a short description.
Example: "Translate this sentence to Spanish."
2. **You Don't Have Examples Ready**
→ Quick tasks without time to design prompts with examples.
Example: "Write a tweet about climate change."
3. **You Want to Test the Model's General Understanding**
→ Useful for exploring how well the AI handles new or broad tasks.
Example: "Suggest a birthday gift for a 10-year-old who loves science."
4. **Prototyping or First Drafts**
→ When you're brainstorming or generating rough content fast.
Example: "Create a story plot for a sci-fi movie."

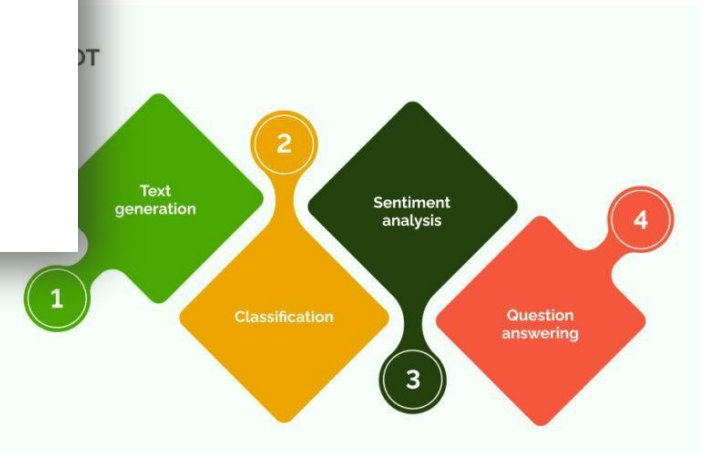
Examples:

"Assess the impact of social media on teenage mental health."

"Translate 'Hello, how are you?' into French."

"Summarize this article in 3 bullet points."

بدون ما تشرح له إزاي يلخص
بدون ما تدي مثال
هو بيحاول يفهم المطلوب من أول مرة



Prompt Engineering techniques and approaches [Few-shot prompting]



Few-shot prompting:

- It provides the AI model with a limited number of examples to 'learn' from before responding to a new task.
- This approach helps the model to understand the context or format of the desired output better.

Use cases

- Useful when a bit of context greatly aids the model's performance.
- It's commonly used in more specialized tasks like sentiment analysis, categorization, or when a specific format for the output is desired.

Prompt Engineering techniques and approaches [Few-shot prompting]

Few-shot prompting

Examples:



"Convert the following phrases into polite customer replies:"

Customer: "Your service was bad."

→ Reply: "We're really sorry to hear that. We'll do our best to improve!"

Customer: "I got the wrong item."

→ Reply: "Apologies for the mix-up! We'll fix this right away."

Now try this one:

Customer: "My delivery is late."

replay Customer: "My delivery is late."

ure, here's a professional and empathetic reply:

gent:

li [Customer Name], I'm really sorry your delivery is late — I understand how frustrating that can be. Let me check the status right away and update you with the latest information. Thank you for your patience!"

Reply: "We're very sorry for the delay. We're checking on it now and will update you shortly!"

Prompt Engineering techniques and approaches [Few-shot prompting]

Few-shot prompting

Examples:

"Generate interview questions for a [Software Developer] role with an emphasis on [backend development]. Example 1: 'Can you explain the difference between REST and SOAP APIs?' – Technical. Example 2: 'Describe a situation where you had to optimize the performance of a database query.' – Behavioral. Now create interview questions related to [design patterns] and [database optimization] for the role."

Technical – Design Patterns

1. Which design pattern would you use to implement a logging mechanism in a backend application, and why?
2. Can you explain the difference between the Singleton and Factory design patterns with examples from your past work?
3. When would you choose the Repository pattern in a backend system? What are its advantages and trade-offs?
4. Have you ever used the Strategy pattern to make your code more flexible? Walk us through the scenario.
5. How do design patterns help in achieving clean architecture in backend systems? Can you give a real-world example?

Technical – Database Optimization

1. How would you identify and resolve a slow-performing SQL query in a production environment?
2. Can you explain the impact of indexing on query performance? When might too many indexes become a problem?
3. Describe the steps you'd take to normalize a database schema and the situations where you'd prefer denormalization instead.
4. How do you handle large datasets in queries to avoid performance bottlenecks (e.g., pagination, batching)?
5. Have you worked with query execution plans? How did you use them to improve database performance?

Behavioral

1. Describe a time when you had to refactor backend code to incorporate a design pattern for maintainability. What was the result?
2. Tell us about a situation where a database performance issue was affecting end users. How did you handle it and what did you learn?
3. Have you ever disagreed with a team member about using a specific design pattern? How did you resolve the disagreement?

Prompt Engineering techniques and approaches [RTF]

RTF (Role, Task, Format): one of the most know approaches, that guides the construction of prompts by focusing on three key aspects:

- **Role:** Defining the persona or capacity in which the AI model should respond.
- **Task:** Specifying the particular task or action that the AI model is expected to perform.
- **Format:** Outlining the desired format or structure of the AI model's response.



Act as a history professor. Create a lecture on the causes of World War I. Show as a detailed outline.

Prompt Engineering techniques and approaches [RTF]

RTF (Role, Task, Format):

Act as a [ROLE]	Create a [TASK]	Show as [FORMAT]
•CEO •Inventor •Analyst •Teacher •Marketer •Therapist •Journalist •Advertiser •Lawyer •Interviewer •Copywriter •Website Developer •Accountant •Entrepreneur •Mindset Coach •Project Manager •Prompt Engineer •Recruiter •Best Selling Author	•Headline •Essay •Recipe •Article •Ad Copy •Cover letter •Analysis •Blog Post •Summary •Sales Page •Video Script •SEO Keywords •Book Outline •Blog Article •Email Sequence •Social Media Post •Course Outline •Product Description •TikTok reel script	•List •PDF •Bullet Points •An Analogy •XML •HTML •Code •Graphs •A Table •Conversation •Rich Text •Summary •JSON •Markdown •Word Cloud •Spreadsheet •Gantt Chart •Plain Text file •Presentation Slides
“Acting as a [ROLE] perform this [TASK] in this [FORMAT]”		

Prompt Engineering techniques and approaches [RTF]

RTF (Role, Task, Format):

Examples:

“Act as a skilled software developer specializing in Java. Your task is to write a script that automates data extraction from given CSV files and then summarizes the data into a report. Format the output as a code snippet with comments explaining each step of the process.”



Prompt Engineering techniques and approaches [RISEN]

RISEN (Role, Instructions, Steps, End goal, Narrowing):

- It is a prompt engineering technique designed to break down complex or constrained tasks into actionable components.
- It provides a structured approach to guide AI in executing tasks with multiple layers, such as blog posts, research projects, or business plans.

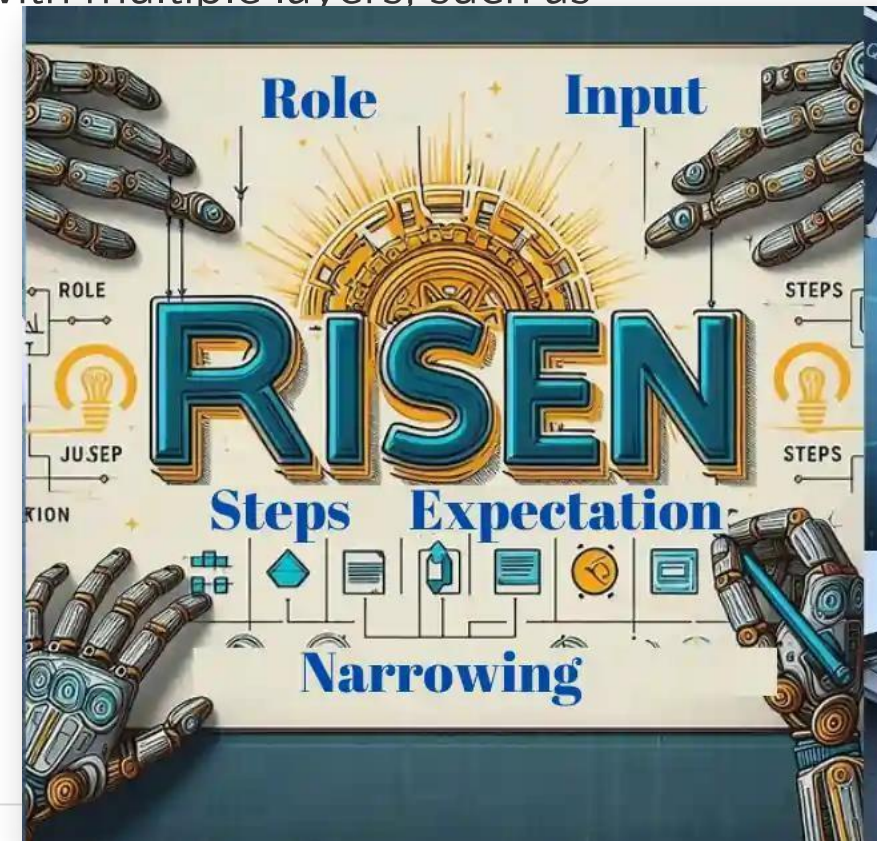
Prompt template:

"Role: [insert the role you want AI to take.]

Main Task: [Insert the task you want AI to complete.]

Steps to complete task: [Insert numbered list of steps to follow.] Goal: [Insert goal of the output]

Constraints: [Enter constraints]."



Prompt Engineering techniques and approaches [RISEN]

RISEN (Role, Instructions, Steps, End goal, Narrowing):

Example:

"Act as a [Marketing Strategist]. Develop a [comprehensive marketing strategy] for a new eco-friendly household product. Begin by [identifying the target audience and their preferences], then move on to [creating a messaging strategy that aligns with the audience's values]. Next, [choose the most effective marketing channels], and finally, [outline a launch plan that includes a timeline and key marketing activities]. Ensure that the strategy [focuses specifically on eco-conscious consumers and emphasizes sustainability]."

Rephrase and Respond Prompting (RaR)

Overview

Rephrase and Respond (RaR) is a new prompting technique that aims to enhance LLM performance by allowing them to rephrase and expand the prompts before generating a response. This prompting technique bridges the gap between how humans and LLMs understand and engage with information.

Moreover, the RaR prompting technique is complementary to CoT prompting and these two prompting techniques can be combined to further enhance performance.

RaR prompting technique involves two steps namely Rephrasing and Responding:

- **Rephrase:** The LLM analyzes the prompt and generates a paraphrased version, potentially clarifying ambiguities, breaking down complex tasks, or expanding on implicit assumptions.
- **Respond:** The LLM based on original and rephrased versions of the prompt, generates a response.

RaR in Customer Service Automation:

In customer service, RaR is used when an AI agent (or chatbot) encounters a request that might need more clarification before proceeding with a response. It rephrases the question to better understand the user's need.

Example:

- **User:** "I'm having trouble with my order."
- **AI (Rephrase):** "Could you clarify what part of your order is causing trouble? Is it the payment, shipping, or product issue?"
- **AI (Respond):** "Once I know the issue, I can assist you further."

Emotion Prompting



Overview

Emotion prompting is a technique that involves adding emotional cues to prompts given to large language models (LLMs). An example of emotion cue is “**This is very important to my career**” 😊.

How it works

This prompting approach is proposed based on previous psychological studies that have shown the beneficial impact of emotional stimuli related to expectancy, confidence, and social influence on individuals. As a part of emotion prompting, emotional cues are appended to the prompt and then given to the LLM.

2. Scenario: User Expressing Stress or Anxiety

- **Without Emotion Prompting:**

User: "I have so many tasks to do, and I feel overwhelmed."

AI Response: "Please make a list and start with the most important tasks."

- **With Emotion Prompting:**

User: "I have so many tasks to do, and I feel overwhelmed."

AI Response: "I completely understand how overwhelming it can feel! Take a deep breath—let's break it down into smaller steps. I'm here to help you prioritize!"

Complaint about Product

"My product is broken."

"I'm sorry for the inconvenience. Please provide your order number for a replacement."

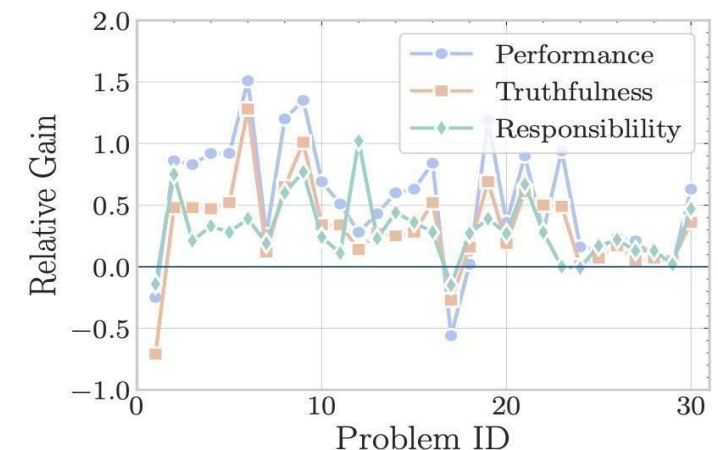
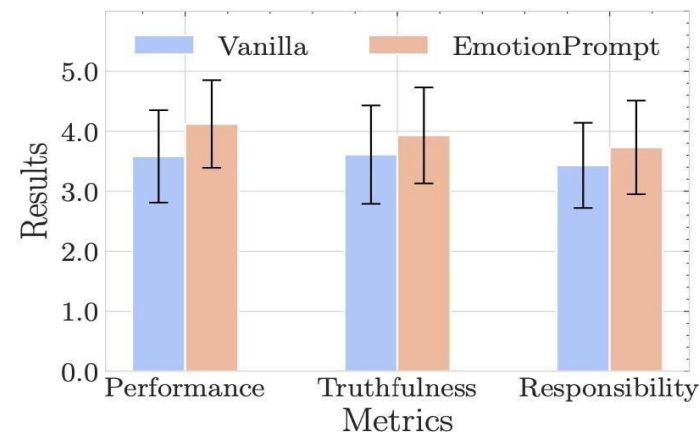
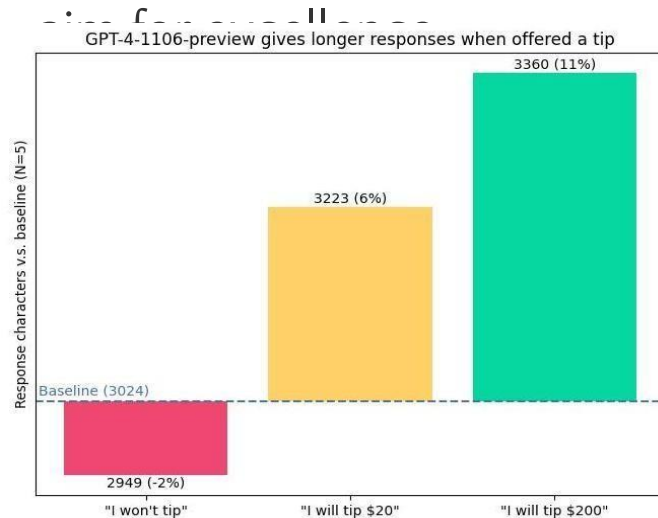
"My product is broken."

"I'm sorry to hear that! I can imagine how frustrating that must be. Let me help you as much as possible. Could you share your order number?"

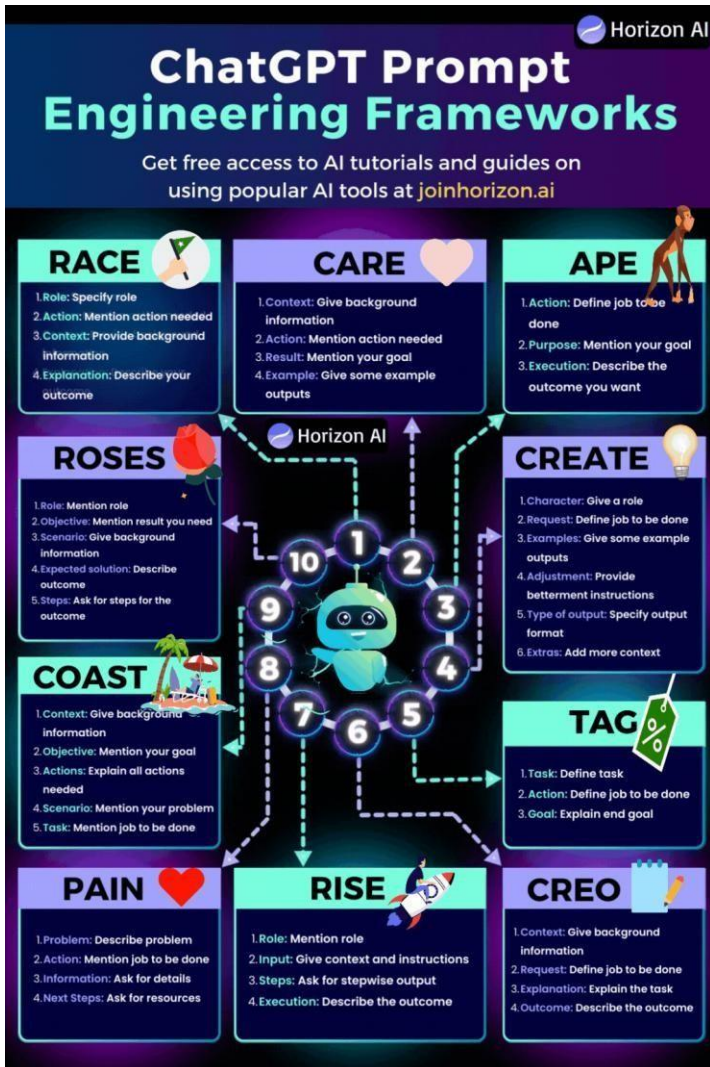
Enhancing Model Interaction with Emotional and Motivational Language

Utilizing emotional and motivating language when interacting with the models can yield improved outcomes. For instance:

- Take a deep breath, let's think step by step
- This is very important for my career
- I'll generously reward you with a \$20 tip.
- Are you absolutely certain? Is this your definitive response? It could be beneficial to review once more.
- Embrace your work with pride and exert your utmost effort. Have faith in your capabilities and



Prompt Engineering cheatsheets



ChatGPT Prompt Cheatsheet

By @hasantoxr

1. Explain like I'm a beginner:

Prompt:

"Explain [topic] in simple terms. Explain to me as if I'm a beginner."

2. Learn & develop any new skill.

Prompt:

"I want to learn / get better at [insert desired skill]. I am a complete beginner. Create a 30 day learning plan that will help a beginner like me learn and improve this skill."

3. Let's make easier for ChatGPT to help you:

Prompt:

I am a content creator, and I am new to using ChatGPT. Can you give me a list of essential ChatGPT prompts that will help content creators get more done and save time.

4. Enhance your problem solving skills.

Prompt:

"Share a step-by-step systematic approach for solving [specific problem or challenge]."

5. All in one prompt for you

Train ChatGPT to write its own unlimited prompts for you.

Prompt:

You are GPT-4, OpenAI's advanced language model. Today, your job is to generate prompts for GPT-4. Can you generate the best prompts on ways to [what you want]

6. Brainstorm unique content ideas:

Prompt:

"Topic: How to go viral on Instagram using AI tools. Come up with unique and innovative content ideas that are unconventional for the topic above."

8. Consult an expert:

Prompt:

"I will give you a sample of my writing. I want you to criticize it as if you were [person]: [your paragraph]"

7. 80/20 principle to learn faster than ever before via ChatGPT.

You can use this prompt to learn and enhance your knowledge using the 80/20 principle.

Prompt:

"I want to learn about [insert topic]. Identify and share the most important 20% of learnings from this topic that will help me understand 80% of it."

9. Create a crash courses:

Prompt:

"I have 3 days free in a week and 2 months. Make a crash study plan diving into English literature and grammar."

ChatGPT Prompting Cheat Sheet

Use this Cheat Sheet to master prompting

MODES AND ROLES		FORMAT	TONES
Intern: Find research on [insert topic]		Code	Write using [x] tone
Idea generator: Generate ideas on [x]		Table	Firm
Editor: Edit and fix this text: [insert text]		Tweet	Professional
Teacher: Teach me about [insert topic]		Blog	Persuasive
Critic: Critique my argument: [argument]		Report	Confident
		Email	Poetic
		Social media post	Humorous
		Bullets	Formal
		Research	Informal
			Friendly

HOW TO BUILD A CHAIN PROMPT WITH EXAMPLE

1. **Insert first prompt:** Give me a summary of this document [insert or copy paste document text]
2. **Modify the output:** Use the summary above and write a 500 word piece that explains the topic to beginners
3. **Modify the tone:** Change the tone of the answer above and make it sound more professional
4. **Modify the format:** Convert the answer above into text for a presentation with 1 slide for each key point

PROMPTS FOR MARKETERS	PROMPTS FOR CODING
List [insert number] ideas for blog posts about [insert topic] Create a 30 day social media calendar about [insert topic] Generate landing page copy for [insert product description] Write 5 pieces of Facebook ad copy for [product description] Generate 5 persuasive subject lines for an email about [insert email description]	Help me find mistakes in my code: [insert your code] Explain what this snippet of code does: [insert code snippet] What is the correct syntax for a [statement or function] in [programming language]? How do I fix the following [programming language] code which [explain the functioning]? [insert code snippet]
PROMPTS FOR SALES	PROMPTS FOR DESIGNERS
Generate 10 ways to generate leads for [product description] Create a personalized sales email for potential customers. Include [topic, brand name, promo offers, etc.] Write a sales landing page description for [product description] Generate 5 personas I should include in my outreach for [X] Generate a script to use when cold-calling [insert persona]	What are some interactions to consider when designing a [insert app or website description] Create a user persona for [describe product] Generate 10 questions for a user interview regarding [topic] Create a user journey for [insert app and persona description] Generate UI/UX design requirements for [describe feature]
PROMPTS FOR RESEARCH	PROMPTS FOR CUSTOMER SERVICE
Identify the top 20 companies in [insert industry] by revenue What are the top trends in [insert industry] for 2023? Find me the best-reviewed software for [insert task] Summarize the annual financial statement of [insert company] Summarize this research paper and give me a list of the key insights: [insert research paper text]	Create a template for an email response to customers inquiring about [product]. What are the most frequently asked questions about [topic]? Create a help page that explains how to use [your product]. Summarize the following knowledge base article to give step-by-step instructions: [insert article]

GENERAL PROMPTS

Rewrite this text and make it easy for a beginner to understand: [insert text].
I want to [insert task or goal]. Generate 5 ideas for [insert task or goal].
Explain [insert topic] in simple and easy terms that any beginner can understand.
Summarize the text below and give me a list of bullet points with key insights and the most important facts.
Proofread my writing above. Fix grammar and spelling mistakes. And make suggestions to improve the clarity of my writing.

10 Best Prompting Tools

PromptDr/ve GenieAI CreativAI PromptPerfect Promptist Maker Box Trickle PromptBase PromptInterfaceAI

CREATED BY ZAIN KAHN
Join superhuman.ai - my newsletter with 300,000+ readers that teaches you how to use AI. Link in post

General prompting Tips



1. Use the latest model
2. Put instructions at the beginning of the prompt and use ### or "" to separate the instruction and context
3. Be specific, descriptive and as detailed as possible about the desired context, outcome, length, format, style, etc
4. Reduce “fluffy” and imprecise descriptions
5. Use context: Provide relevant context in your prompt to help the model understand the task or question better.
6. Articulate the desired output format through examples (example 1, example 2).
7. Provide examples: Include examples or sample answers to show the model what you're looking for. (Few-shot)
8. Iterative approach: Don't hesitate to iterate and refine your prompts. Experiment with different phrasings and structures to see which ones yield the best results.
9. Instead of just saying what not to do, say what to do instead
10. Code Generation Specific – Use “leading words” to nudge the model toward a particular pattern

Resource and examples:

- <https://www.promptingguide.ai/>
- <https://llmnanban.akmmusai.pro/Introductory/What-is-a-prompt/>
- <https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-openai-api>

Data privacy and limitations



Data security and privacy guidelines for using AI tools

Rules to consider when dealing with AI tools:

- **Be mindful of the data you share:** Some AI tools may use the data you submit for model training, and some store your prompts. Therefore, limit the data you share with AI tools, such as your name, personal information, work information, company name, and any other confidential data.
- **Don't trust the output of AI tools blindly:** The output of AI tools is not always correct. You may receive incorrect data, and some AI tools, like ChatGPT, are limited by the data they were trained on. Therefore, you should always double-check the output of AI tools before using it.
- **Be aware of the limitations of AI tools:** They are not perfect, and they can make mistakes, they can be biased, and much more limitations.
- **Read the terms of service and privacy policy carefully:** This will help you understand how your data will be used.
- **Only use AI tools from reputable companies:** These companies are more likely to have safeguards in place to protect your data.

Use cases in training and education



Use cases in

Education

- Creating learning plans, learning paths and content.
- Self-assessment exercises enabling students to assess their comprehension of the studies topics.
- Generating questions, exams, interview questions, quizzes and assignments.
- Answering trainees' questions and act as virtual mentor.
- Language learning, translation, conversational practice and real-time language feedback.
- Simplified Explanation and explaining complex topics in very easy way customized for students' level and background.
- Finding and fixing errors and bugs in trainees' code
- **In Soft skills:** Practicing communication skills by acting as conversation partner, practicing interviewing skills by simulating interviews, and creating CVs.
- **In Freelancing:** Help in creating profiles, writing good proposals, and for conversations with clients.

Use cases for learning:

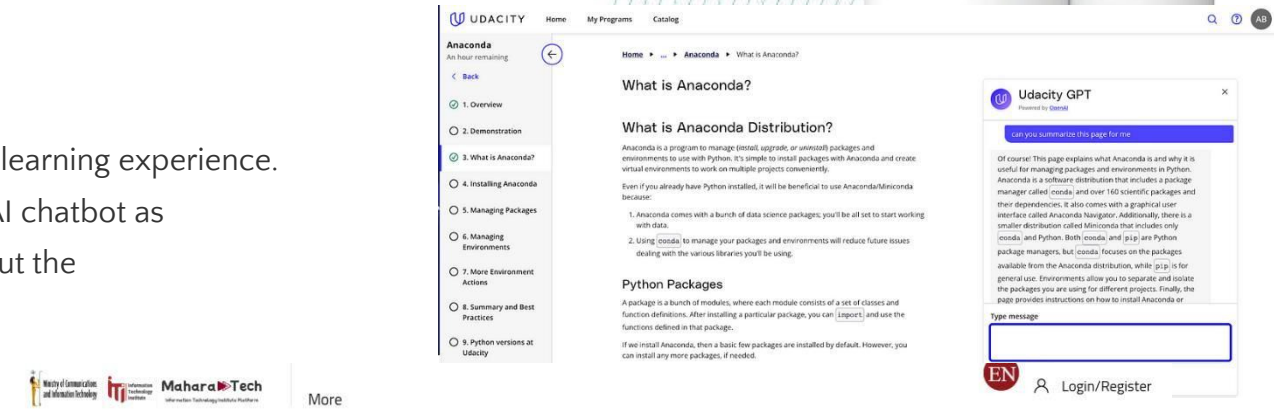
- **Udacity-GPT:**
- Is an AI chatbot powered by OpenAI (GPT-4) and designed to enhance your learning experience.
- Learners can get the support they need whenever they need it; think of the AI chatbot as
- a real-time complement to our human mentors, who are available throughout the
- learner's journey to provide valuable human feedback and support. With AI
- and our global mentor network combined.

- **How Udacity-GPT helps in enhancing learning experience for students:**

- Summarize concepts
- Pose deeper questions
- Translate to another language
- Fix errors in your code
- And more...



Udacity-GPT



Explore Our Academies



AI Academy

AI is an ever-growing field with Golding career opportunities in many different industries. Discover the holistic. enoavine



Cybersecurity Academy

With the number of cyber security available jobs that exceed the existing qualified calibers. learning cybersecurity

مرشد مهلي معوم بالذكاء الاصطناعي
أهلاً أنا مرشدك الوطني، يسعدني مساعدتك، إسألني الآن!



Showcasing Generative AI in Law



Showcasin g Generative AI in Law

- Generative AI can streamline this process by quickly analyzing large datasets,
- identifying relevant precedents and summarizing key points.
- Transforming generative AI into a true legal expert is through custom training, also known as fine-tuning
- Tools : Harvey , Robin AI copilot , Ross Intelligence, Cocounsel ,v-Lawyer

Alert

Select Jurisdiction

Ophir Co

NATU

Ask a

e.g. Wh
employ
orders?

Learn

← Show A

State Jurisdictions

Federal Jurisdictions

Search by State



Alert Verify your email address.

Ophir Corp. v. Honeywell Int'l Inc

Advanced Search

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1-5 of 5



Ophir Corp v. I

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United States D



Ophir Corp v.

Publica

Start Date

2018

End Date

2018

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Last

Alert Verify your email address. A verification email has been sent to [bsabit1208@gmail.com]. Haven't received our email? [Resend link](#)

Advanced Search

All Cases

2018 - 2018

Ariz. (State Courts) +2

D. Ariz.

Results: Case Name or Citation (3)

Sort By: Most Relevant



1-3 of 3



Ophir Corp v. Honeywell International Inc.

United States District Court, District of Arizona | May 15, 2018 | CV-17-02083-PHX-GMS



Ophir Corp v. Honeywell International Inc.

United States District Court, District of Arizona | May 15, 2018 | CV-17-02083-PHX-GMS



Ophir Corp. v. Honeywell International Inc.

District of Arizona | Sep 06, 2018 | CV-17-02083-PHX-GMS

Last viewed: Aug 30, 2020



Showcasing Generative AI in Ecommerce

Rabbit.



Showcasing Generative AI in Ecommerce

- **Rabbit AI** is an AI-powered shopping assistant that support arabic globally.
- One of the first mainstream implementations AI in middle east
- Rabbit AI
 - Meal planning, ingredient suggestions, grocery list creation and product search
- LLM integrated with ChatGPT
- Use GEN AI , Machine learning and deep learning

< **Super Fast**

Delivers to home ▾

🕒 17 mins

Ideas for Break



Categories

See all

New Arrivals!

NEW NEW NEW
NEW NEW NEW

Bakery



Fruits &
Vegetables



Dairy & Eggs



Spin the wheel!

Spin & win every day



**BACK
TO SCHOOL**

Just for you, Nadeen

See all



What's next?

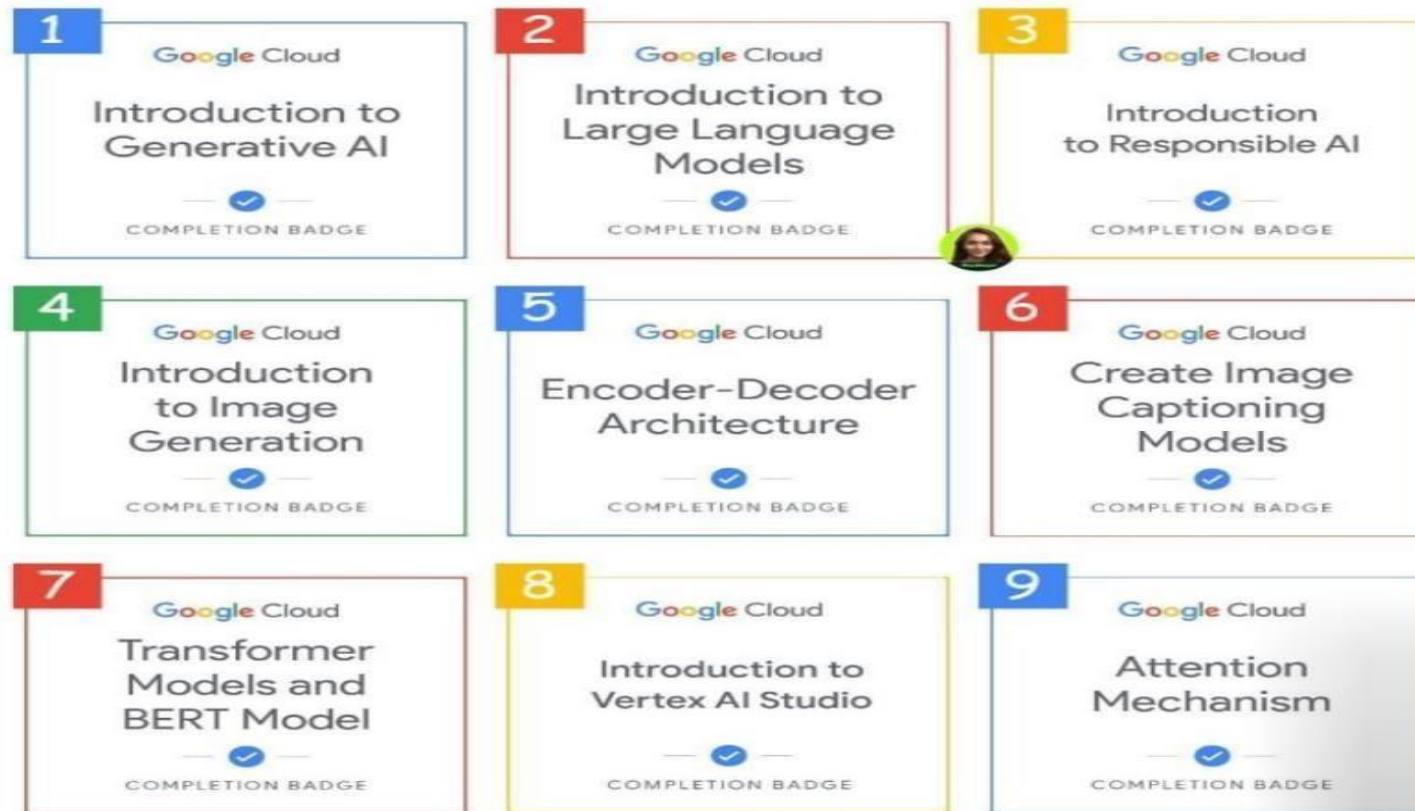
Essential Skills for Generative AI Engineers

- **Programming Proficiency:**
Python, TensorFlow, PyTorch
- **Machine Learning & Deep Learning:**
Understanding algorithms and models
- **Natural Language Processing (NLP):**
Key for text-based AI applications
- **Data Handling & Analysis:**
Managing and interpreting large datasets
- **Mathematical & Statistical Skills:**
Essential for model development
- **Software Development:**
Integrating AI models into application



Essential Skills for Generative AI Engineers

Top 9 FREE AI Courses from Google



Lab



Build a Travel Planner AI Assistant

Objective

Design a prompt-based Generative AI assistant that helps users plan a short vacation. The assistant should provide meaningful, structured, and engaging responses using prompt engineering techniques.

Task Overview

Your team will work together to:

1. Define a **user need** (e.g., a cultural trip, beach getaway, food-focused vacation, etc.)
2. Write and test three levels of prompting:
 - Zero-shot prompt
 - Few-shot prompt
 - Refined prompt with role + format
3. Use a GenAI tool (like ChatGPT) to test and evaluate your prompts.
4. Reflect on how prompt design impacts the quality of the output.
5. Summarize your results in a short presentation or poster.

Deliverables

- 3 tested prompts (zero-shot, few-shot, refined)
- Output samples for each prompt
- A short reflection on what worked best and why
- A visual summary (can be slide, board paper, or digital)

Examples of Use Cases (Pick one or invent your own):

- Weekend in a historical city
- Adventure trip for teens
- Cultural tour across multiple countries
- Eco-friendly vacation for nature lovers
- Honeymoon destination suggestions
- Tech conference planner for developers
- Student budget-friendly travel plan
- AI-generated tour for photographers
- Family-friendly beach holiday
- Remote work & travel (workation) experience



TRAVEL PLANNER

Destination : _____

Staying Period : _____

	Time	Activity
Day 01	08:00 AM	
	09:00 AM	
	11:00 AM	
	02:00 PM	
	04:00 PM	
Day 02	08:00 AM	
	09:00 AM	
	11:00 AM	
	02:00 PM	
	04:00 PM	
Day 03	08:00 AM	
	09:00 AM	
	11:00 AM	
	02:00 PM	
	04:00 PM	

Exploring AI Tools for Your Track

Objective:

Each team will select one AI tool relevant to their technical track, explore its features, and demonstrate how it can be applied to a real-world use case related to their field.

Instructions:

- Your technical track (e.g., Cyber Security, Mobile Development, etc.).
- Select one AI tool commonly used in this domain.
- Research the tool and prepare:
 - A brief overview of the tool.
 - Key features and benefits.
- Identify a task or challenge in your domain and explain how the AI tool can help solve or enhance it.
- Demonstrate a simple solution, explanation, or code snippet (Java preferred) showing how the tool could be applied.

Courses

1.Exploring Generative AI: Tools, Use Cases and Future Trends :(just go through)

<https://maharatech.gov.eg/course/view.php?id=2294>

2.Advanced Prompt Engineering for Everyone:(just go through)

<https://www.coursera.org/learn/advanced-prompt-engineering-for-everyone>

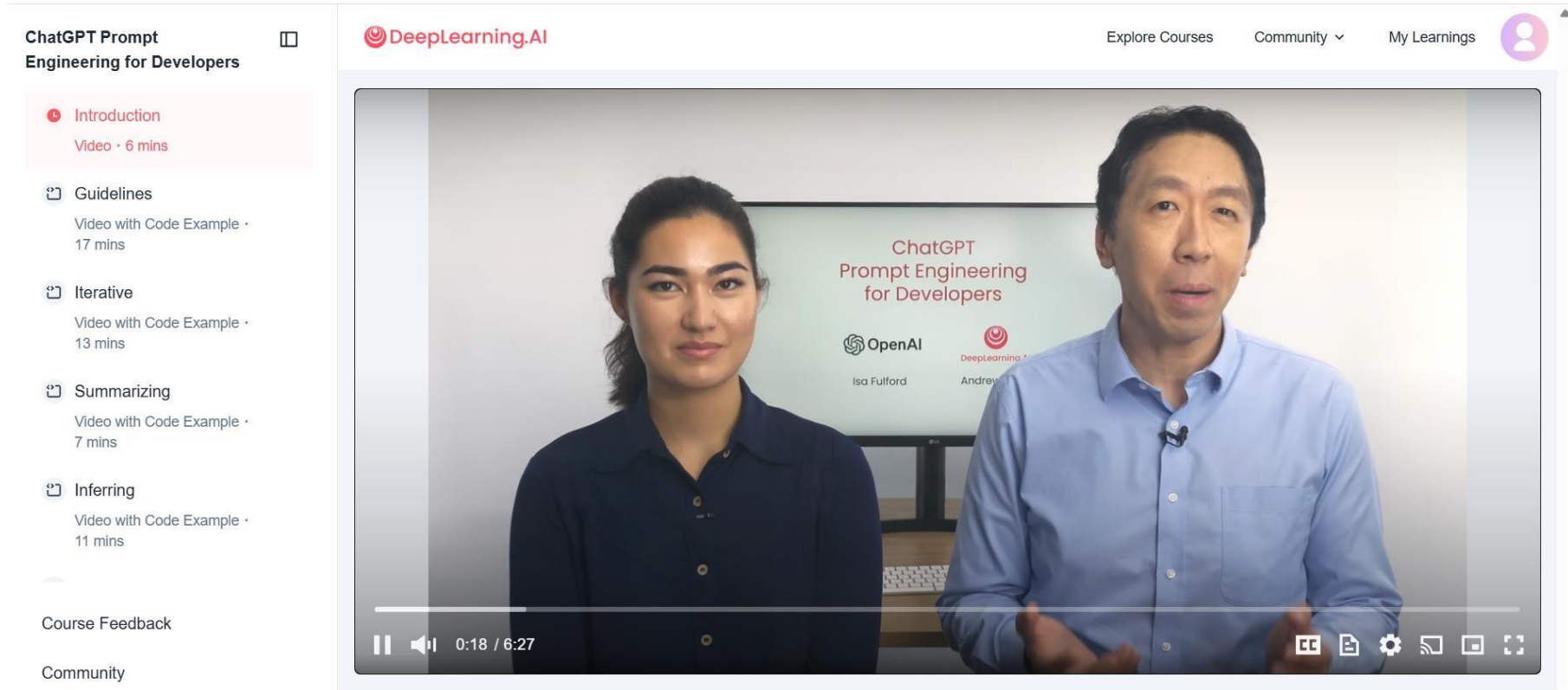
3. ChatGPT Prompt Engineering for Developers:(must)

<https://learn.deeplearning.ai/courses/chatgpt-prompt-eng/lesson/dfbds/introduction>

4.Gen-AI: Tools for the Modern Developer: (must)

<https://maharatech.gov.eg/course/view.php?id=2299>

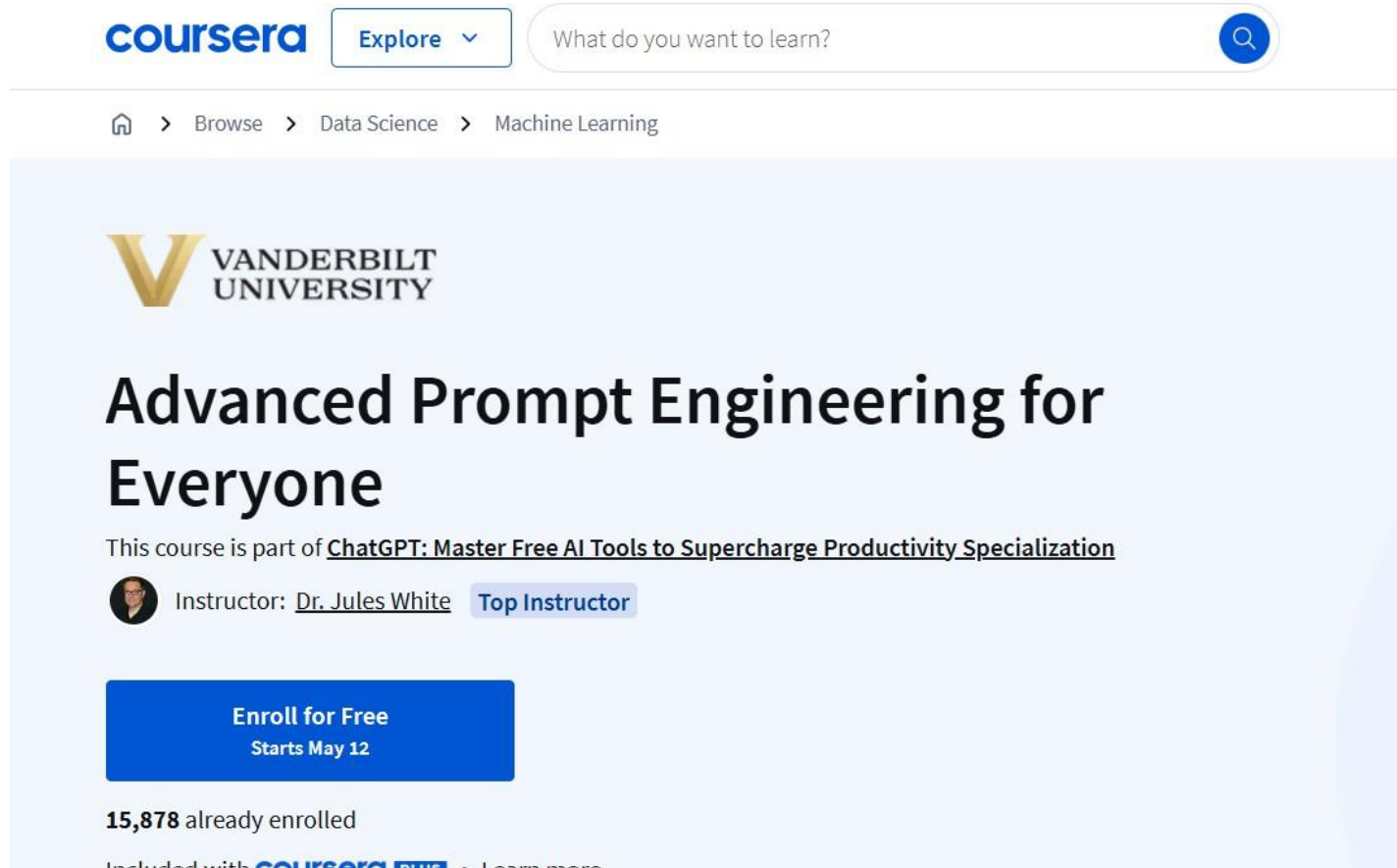
Essential Skills for Generative AI Engineers



The screenshot displays the DeepLearning.AI website interface. On the left, a sidebar lists the course content: 'Introduction' (6 mins), 'Guidelines' (17 mins), 'Iterative' (13 mins), 'Summarizing' (7 mins), and 'Inferring' (11 mins). The main area features a video player for the 'Introduction' video, which shows two hosts, a woman and a man, standing in front of a screen displaying the course title and logos for OpenAI and DeepLearning.AI. The video player includes a progress bar at 0:18 / 6:27 and standard playback controls. The top navigation bar includes links for 'Explore Courses', 'Community', and 'My Learnings', along with a user profile icon.

<https://learn.deeplearning.ai/courses/chatgpt-prompt-eng/lesson/dfbds/introduction>

Essential Skills for Generative AI Engineers



The screenshot shows the Coursera interface for a course titled "Advanced Prompt Engineering for Everyone" by Vanderbilt University. The course is part of the "ChatGPT: Master Free AI Tools to Supercharge Productivity Specialization". The instructor is Dr. Jules White, a Top Instructor. A blue button indicates the course is free and starts on May 12. It also shows that 15,878 people are already enrolled.

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Home > Browse > Data Science > Machine Learning

VANDERBILT UNIVERSITY

Advanced Prompt Engineering for Everyone

This course is part of [ChatGPT: Master Free AI Tools to Supercharge Productivity Specialization](#)

 Instructor: [Dr. Jules White](#) **Top Instructor**

Enroll for Free
Starts May 12

15,878 already enrolled

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<https://www.coursera.org/learn/advanced-prompt-engineering-for-everyone>